

# Supplement to “Multiscale Comparison of Nonparametric Trend Curves”

## A Simulations

In this section, we investigate our test and clustering methods by means of Monte Carlo experiments.

### A.1 Size and power properties of the multiscale test

The simulation design is as follows: We generate data from the model  $Y_{it} = m_i(\frac{t}{T}) + \beta_i^\top \mathbf{X}_{it} + \alpha_i + \varepsilon_{it}$ , where the number of time series is set to  $n = 15$  and we consider different time series lengths  $T$ . We include 3 covariates and model them by the VAR(3) process

$$\mathbf{X}_{it} = \begin{pmatrix} a_1 & 0 & 0 \\ 0 & a_2 & 0 \\ 0 & 0 & a_3 \end{pmatrix} \mathbf{X}_{i,t-1} + \boldsymbol{\nu}_{it},$$

where  $\mathbf{X}_{it} = (X_{it,1}, X_{it,2}, X_{it,3})^\top$ ,  $\boldsymbol{\nu}_{it} = (\nu_{it,1}, \nu_{it,2}, \nu_{it,3})^\top$  and we choose  $a_1 = a_2 = a_3 = 0.25$ .

The innovations  $\boldsymbol{\nu}_{it}$  are drawn i.i.d. from a multivariate normal distribution  $N(0, \Phi)$  with

$$\Phi = \begin{pmatrix} 1 & \varphi & \varphi \\ \varphi & 1 & \varphi \\ \varphi & \varphi & 1 \end{pmatrix},$$

where  $\varphi = 0.25$  specifies the correlation between the 3 covariates. The parameter vector  $\beta_i = (\beta_{i,1}, \beta_{i,2}, \beta_{i,3})^\top$  is chosen to equal  $(1, 1, 1)^\top$  for each  $i$ . Moreover, for each  $i$ , the idiosyncratic error process  $\{\varepsilon_{it} : 1 \leq t \leq T\}$  follows the AR(1) model  $\varepsilon_{it} = a\varepsilon_{it-1} + \eta_{it}$ , where  $a = 0.25$  and the innovations  $\eta_{it}$  are i.i.d. normally distributed with mean 0 and standard deviation 0.25. The vector of fixed effects  $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_n)^\top$  is modelled as a

normally distributed random vector. In particular,  $\boldsymbol{\alpha} \sim N(0, \Sigma)$  with

$$\Sigma = \begin{pmatrix} 1 & \rho & \cdots & \rho \\ \rho & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \rho \\ \rho & \cdots & \rho & 1 \end{pmatrix},$$

where  $\rho = 0.25$  gives the correlation across  $i$ . To generate data under the null  $H_0 : m_1 = \dots = m_n$ , we let  $m_i = 0$  for all  $i$ . (A robustness check with a different choice of  $m_i$  is discussed below.) To produce data under the alternative, we use the bump function

$$m_1(u) = b \cdot \vartheta(u, 0.3, 0.1) - b \cdot \vartheta(u, 0.7, 0.1) \quad (\text{A.1})$$

with  $\vartheta(u, c, d) = \mathbb{1} \left\{ \frac{|u-c|}{d} \leq 1 \right\} \left( 1 - \frac{(u-c)^2}{d^2} \right)^2$  for  $b \in \{0.25, 0.5, 0.75\}$  (depicted in Figure A.1) and let  $m_i = 0$  for  $i \neq 1$ . Note that the normalisation constraint  $\int_0^1 m_1(u) du = 0$  is directly satisfied in this case. For each simulation exercise, we draw 5000 samples.

Figure A.1 illustrates that the described simulation setup is quite challenging already without covariates and fixed effects. The black lines in the plot show the bump function  $m_1$  for different heights of the bump  $b \in \{0, 0.25, 0.5, 0.75\}$ , where the value  $b = 0$  corresponds to the null  $H_0$ . The red lines show one realization of the time series  $\{Y_{1t}^\circ : 1 \leq t \leq T\}$  for  $T = 250$ , where  $Y_{1t}^\circ = m_1(\frac{t}{T}) + \varepsilon_{1t}$  is an idealized version of  $Y_{1t}$  without covariates and fixed effects. The values  $b = 0.25, 0.5, 0.75$  correspond to three different specifications of  $m_1$  under the alternative. In particular, the larger the bump size  $b$ , the further away  $m_1$  is from the other functions  $m_i, i \neq 1$ . For the smallest value  $b = 0.25$ , the bump function is very difficult to detect since the noise level is very high, or put differently, since the signal-to-noise ratio is very small (as is clearly visible in the upper right panel of Figure A.1). For  $b = 0.5$  and  $b = 0.75$ , the signal-to-noise ratio slowly gets bigger, making it less difficult to detect the bump function. Nevertheless, even for  $b = 0.75$ , there is substantial noise around the bump function.

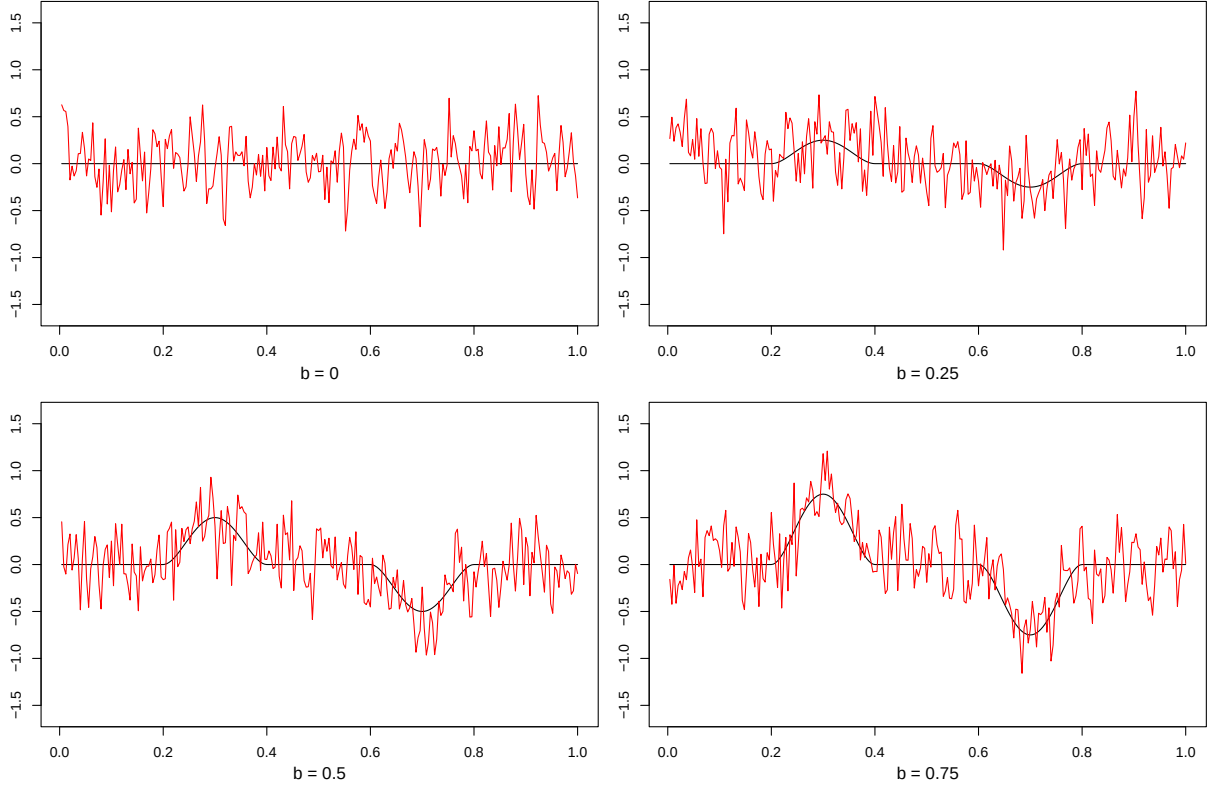


Figure A.1: In black: the bump function  $m_1$  for different bump heights  $b \in \{0, 0.25, 0.5, 0.75\}$  with  $b = 0$  corresponding to the null  $H_0$ . In red: the time series  $\{Y_{1t}^o : 1 \leq t \leq T\}$  with  $Y_{1t}^o = m_1(t/T) + \varepsilon_{1t}$  for one simulation run and  $T = 250$ .

Our multiscale test is implemented as follows: The estimators  $\hat{\beta}_i$  and  $\hat{\alpha}_i$  of the unknown parameters  $\beta_i$  and  $\alpha_i$  are computed as described in Section 3.1. The long-run error variance is estimated by the difference-based estimator proposed in Khismatullina and Vogt (2020), where the tuning parameters  $q$  and  $r$  are set to 25 and 10, respectively. In order to construct our test statistics, we use an Epanechnikov kernel and the location-scale grid  $\mathcal{G}_T = U_T \times H_T$  with

$$U_T = \{u \in [0, 1] : u = \frac{5t}{T} \text{ with } 1 \leq t \leq T\} \quad (\text{A.2})$$

$$H_T = \{h \in [\frac{\log T}{T}, \frac{1}{4}] : h = \frac{5t-3}{T} \text{ with } 1 \leq t \leq T\}. \quad (\text{A.3})$$

We thus consider intervals  $\mathcal{I}_{u,h} = [u - h, u + h]$  which contain 5, 15, 25, ... data points. The critical value  $q_{n,T}(\alpha)$  of our test is computed by Monte Carlo methods as described in Section 3.4, where we set  $L = 5000$ .

Table A.1: Empirical size and power of the multiscale test for different time series lengths  $T$  and nominal sizes  $\alpha$ . Each panel gives the results for a different bump height  $b$ , where  $b = 0$  corresponds to the null  $H_0$  and  $b = 0.25, 0.5, 0.75$  to three different alternatives.

| (a) $b = 0$           |       |       |       | (b) $b = 0.25$        |       |       |       |
|-----------------------|-------|-------|-------|-----------------------|-------|-------|-------|
| nominal size $\alpha$ |       |       |       | nominal size $\alpha$ |       |       |       |
| $T$                   | 0.01  | 0.05  | 0.1   | $T$                   | 0.01  | 0.05  | 0.1   |
| 100                   | 0.006 | 0.035 | 0.070 | 100                   | 0.011 | 0.046 | 0.095 |
| 250                   | 0.020 | 0.079 | 0.134 | 250                   | 0.158 | 0.348 | 0.473 |
| 500                   | 0.012 | 0.058 | 0.112 | 500                   | 0.524 | 0.762 | 0.854 |
| (c) $b = 0.5$         |       |       |       | (d) $b = 0.75$        |       |       |       |
| nominal size $\alpha$ |       |       |       | nominal size $\alpha$ |       |       |       |
| $T$                   | 0.01  | 0.05  | 0.1   | $T$                   | 0.01  | 0.05  | 0.1   |
| 100                   | 0.012 | 0.058 | 0.105 | 100                   | 0.010 | 0.035 | 0.071 |
| 250                   | 0.837 | 0.952 | 0.979 | 250                   | 0.967 | 0.992 | 0.996 |
| 500                   | 1.000 | 1.000 | 1.000 | 500                   | 1.000 | 1.000 | 1.000 |

Table A.1 reports the empirical size and power of the multiscale test. Both the empirical size and the empirical power are computed as the number of simulations in which the test rejects the global null  $H_0$  divided by the total number of simulations. The four panels (a)–(d) of Table A.1 correspond to different values of the bump size  $b$ . Panel (a) corresponds to the case  $b = 0$  and thus gives the empirical size of the test under the null  $H_0$ . Panels (b)–(d), in contrast, report the empirical power against the three alternatives with  $b \in \{0.25, 0.5, 0.75\}$ . Inspecting panel (a), one can see that the empirical size of the test gives a reasonable approximation to the target  $\alpha$  in all scenarios under investigation. Inspecting panels (b)–(d), one can further see that the test has good power properties. For the smallest time series length  $T = 100$ , the power numbers are still very low, reflecting the fact that the noise level in the data is extremely high, even for the largest bump size  $b = 0.75$ . However, as we increase the time series length  $T$ , the power increases quickly. For  $T = 250$  and  $b = 0.5$ , for instance, we already have power of over 95% when  $\alpha \geq 0.05$ .

The empirical power computed in Table A.1 is a very rough measure of performance. We now refine it to get more insight into the performance of our test methods. To do so, we

introduce the following notation. We say that our test finds

- a **real difference** between  $i$  and  $j$  on the interval  $[u - h, u + h]$  if the local null hypothesis  $H_0^{[i,j]}(u, h)$  is violated (i.e., if the trends  $m_i$  and  $m_j$  are indeed different on  $[u - h, u + h]$ ),
- a **spurious difference** between  $i$  and  $j$  on  $[u - h, u + h]$  if  $H_0^{[i,j]}(u, h)$  is not violated (i.e., if the trends  $m_i$  and  $m_j$  are the same on  $[u - h, u + h]$ ).

Notably, the power numbers in Table A.1 do not give any information on (i) whether the differences found by the test are real or spurious and (ii) which real differences have been found, in particular, whether all or only part of them have been found. We now introduce refined notions of power which give this kind of information. The simulation setup is the same as before. In particular,  $m_1$  is the bump function from (A.1) with  $b \in \{0.25, 0.5, 0.75\}$  and  $m_i = 0$  for all  $i \neq 1$ . Hence,  $m_1$  differs from the other trends only in the time period  $\mathcal{I}^* = [0.2, 0.4] \cup [0.6, 0.8]$ . We compute the following notions of power in this setup:

- **real power**<sup>[1]</sup>: the percentage of simulation runs in which our test finds a real difference between  $m_1$  and *at least one* other trend  $m_i$  [i.e., a difference between  $m_1$  and another  $m_i$  on an interval  $[u - h, u + h]$  with  $[u - h, u + h] \cap \mathcal{I}^* \neq \emptyset$ ].
- **real power**<sup>[50]</sup>: the percentage of simulation runs in which our test finds a real difference between  $m_1$  and *at least 50%* of the other trends  $m_i$  [i.e., a difference between  $m_1$  and  $m_i$  on an interval  $[u - h, u + h]$  with  $[u - h, u + h] \cap \mathcal{I}^* \neq \emptyset$  for at least 50% of the other trends  $i$ ].
- **real power**<sup>[100]</sup>: the percentage of simulation runs in which our test finds a real difference between  $m_1$  and *all* other trends  $m_i$  [i.e., a difference between  $m_1$  and  $m_i$  on some interval  $[u - h, u + h]$  with  $[u - h, u + h] \cap \mathcal{I}^* \neq \emptyset$  for all  $i \neq 1$ ].
- **spurious power**: the percentage of simulation runs in which our test finds at least one spurious difference.

Table A.2: Real power<sup>[1]</sup> of the multiscale test.

| (a) $b = 0.25$        |       |       |       | (a) $b = 0.5$         |       |       |       | (b) $b = 0.75$        |       |       |       |
|-----------------------|-------|-------|-------|-----------------------|-------|-------|-------|-----------------------|-------|-------|-------|
| nominal size $\alpha$ |       |       |       | nominal size $\alpha$ |       |       |       | nominal size $\alpha$ |       |       |       |
| $T$                   | 0.01  | 0.05  | 0.1   | $T$                   | 0.01  | 0.05  | 0.1   | $T$                   | 0.01  | 0.05  | 0.1   |
| 100                   | 0.004 | 0.019 | 0.032 | 100                   | 0.004 | 0.016 | 0.034 | 100                   | 0.000 | 0.001 | 0.003 |
| 250                   | 0.130 | 0.285 | 0.392 | 250                   | 0.822 | 0.941 | 0.972 | 250                   | 0.959 | 0.990 | 0.995 |
| 500                   | 0.548 | 0.776 | 0.843 | 500                   | 1.000 | 1.000 | 1.000 | 500                   | 1.000 | 1.000 | 1.000 |

Table A.3: Real power<sup>[50]</sup> of the multiscale test.

| (a) $b = 0.25$        |       |       |       | (a) $b = 0.5$         |       |       |       | (b) $b = 0.75$        |       |       |       |
|-----------------------|-------|-------|-------|-----------------------|-------|-------|-------|-----------------------|-------|-------|-------|
| nominal size $\alpha$ |       |       |       | nominal size $\alpha$ |       |       |       | nominal size $\alpha$ |       |       |       |
| $T$                   | 0.01  | 0.05  | 0.1   | $T$                   | 0.01  | 0.05  | 0.1   | $T$                   | 0.01  | 0.05  | 0.1   |
| 100                   | 0.000 | 0.000 | 0.000 | 100                   | 0.000 | 0.000 | 0.000 | 100                   | 0.000 | 0.000 | 0.000 |
| 250                   | 0.001 | 0.008 | 0.018 | 250                   | 0.260 | 0.497 | 0.644 | 250                   | 0.741 | 0.897 | 0.950 |
| 500                   | 0.050 | 0.150 | 0.226 | 500                   | 0.997 | 1.000 | 1.000 | 500                   | 1.000 | 1.000 | 1.000 |

Table A.4: Real power<sup>[100]</sup> of the multiscale test.

| (a) $b = 0.25$        |       |       |       | (a) $b = 0.5$         |       |       |       | (b) $b = 0.75$        |       |       |       |
|-----------------------|-------|-------|-------|-----------------------|-------|-------|-------|-----------------------|-------|-------|-------|
| nominal size $\alpha$ |       |       |       | nominal size $\alpha$ |       |       |       | nominal size $\alpha$ |       |       |       |
| $T$                   | 0.01  | 0.05  | 0.1   | $T$                   | 0.01  | 0.05  | 0.1   | $T$                   | 0.01  | 0.05  | 0.1   |
| 100                   | 0.000 | 0.000 | 0.000 | 100                   | 0.000 | 0.000 | 0.000 | 100                   | 0.000 | 0.000 | 0.000 |
| 250                   | 0.000 | 0.000 | 0.000 | 250                   | 0.012 | 0.045 | 0.086 | 250                   | 0.215 | 0.428 | 0.568 |
| 500                   | 0.000 | 0.004 | 0.007 | 500                   | 0.782 | 0.927 | 0.960 | 500                   | 1.000 | 1.000 | 1.000 |

Table A.5: Spurious power of the multiscale test.

| (a) $b = 0.25$        |       |       |       | (a) $b = 0.5$         |       |       |       | (b) $b = 0.75$        |       |       |       |
|-----------------------|-------|-------|-------|-----------------------|-------|-------|-------|-----------------------|-------|-------|-------|
| nominal size $\alpha$ |       |       |       | nominal size $\alpha$ |       |       |       | nominal size $\alpha$ |       |       |       |
| $T$                   | 0.01  | 0.05  | 0.1   | $T$                   | 0.01  | 0.05  | 0.1   | $T$                   | 0.01  | 0.05  | 0.1   |
| 100                   | 0.000 | 0.000 | 0.000 | 100                   | 0.010 | 0.035 | 0.071 | 100                   | 0.000 | 0.000 | 0.000 |
| 250                   | 0.000 | 0.000 | 0.000 | 250                   | 0.016 | 0.060 | 0.110 | 250                   | 0.000 | 0.000 | 0.000 |
| 500                   | 0.000 | 0.001 | 0.002 | 500                   | 0.014 | 0.060 | 0.114 | 500                   | 0.000 | 0.001 | 0.001 |

The results for real power<sup>[1]</sup>, real power<sup>[50]</sup>, real power<sup>[100]</sup> and spurious power are presented in Tables A.2–A.5. The real power numbers in Tables A.2–A.4 demonstrate that our test is very capable of finding the real differences between the trends. As can be seen, the numbers increase very quickly as either the sample size  $T$  or the signal strength  $b$  gets larger. In particular, real power<sup>[1]</sup> (Table A.2) exceeds 82% for all nominal sizes  $\alpha$  already for moderate sample sizes  $T \geq 250$  and moderate signals  $b \geq 0.5$ , and approaches 100% for

Table A.6: Empirical size of the multiscale test under  $H_0$  with (a)  $m_i = 0$  for all  $i$  and (b)  $m_i$  the bump function from (A.1) with  $b = 0.25$  for all  $i$ .

| (a) $m_i = 0$ for all $i$ |       |       |       | (b) $m_i$ bump function with $b = 0.25$ for all $i$ |       |       |       |
|---------------------------|-------|-------|-------|-----------------------------------------------------|-------|-------|-------|
| nominal size $\alpha$     |       |       |       | nominal size $\alpha$                               |       |       |       |
| $T$                       | 0.01  | 0.05  | 0.1   | $T$                                                 | 0.01  | 0.05  | 0.1   |
| 100                       | 0.006 | 0.035 | 0.070 | 100                                                 | 0.001 | 0.005 | 0.014 |
| 250                       | 0.020 | 0.079 | 0.134 | 250                                                 | 0.006 | 0.031 | 0.068 |
| 500                       | 0.012 | 0.058 | 0.112 | 500                                                 | 0.011 | 0.049 | 0.093 |

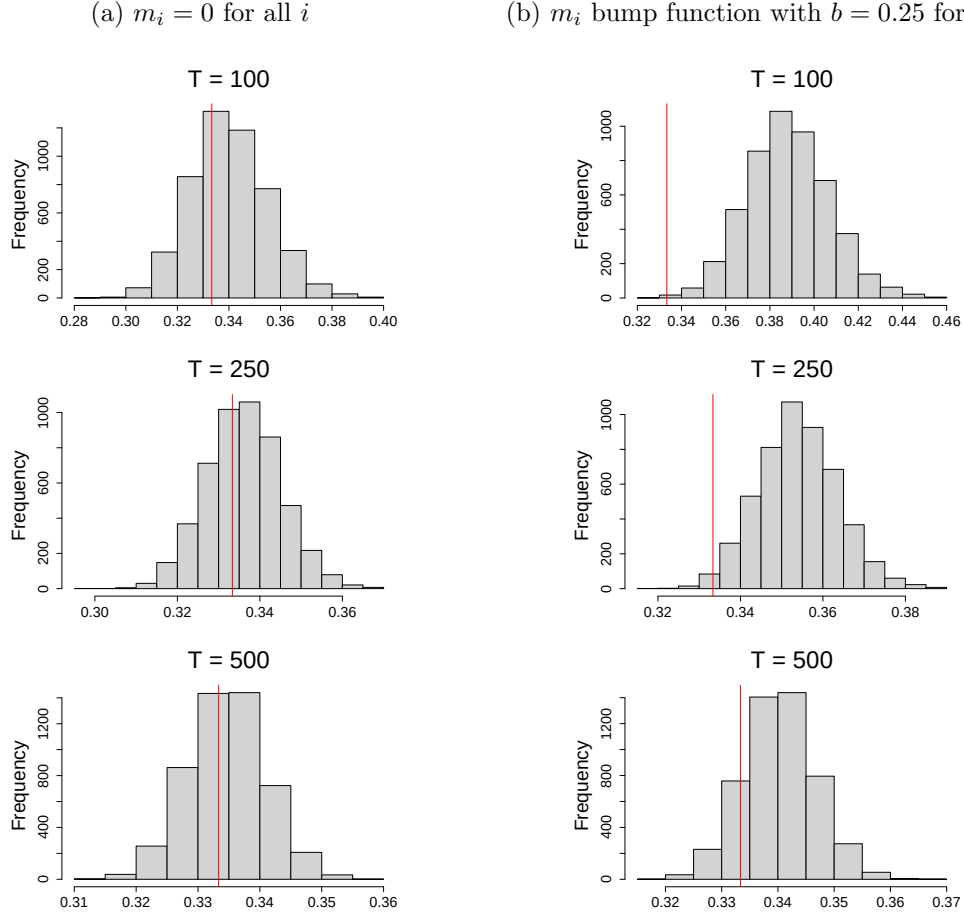
larger  $T$  or stronger signals. Real power<sup>[50]</sup> (Table A.3) and real power<sup>[100]</sup> (Table A.4) are naturally more demanding but still exceed 78% for all nominal sizes  $\alpha$ ,  $T = 500$  and  $b \geq 0.5$ . In addition, the spurious power numbers in Table A.5 demonstrate that the test controls the probability of finding a false positive at the nominal level  $\alpha$ . Taken together, the results in Tables A.2–A.5 illustrate the main advantage of our test: it provides information on which trends are different and where they differ from each other.

## A.2 Robustness checks for the multiscale test

### A.2.1 Additional size simulations

As a first robustness check, we re-run the size simulations from Section A.1 with a different choice of the function  $m_i$ . In particular, rather than choosing  $m_i = 0$  for all  $i$ , we choose  $m_i$  to be the bump function from (A.1) with  $b = 0.25$ . The resulting size numbers are presented in Table A.6(b). For ease of comparison, we have reproduced the size numbers from Table A.1(a) for the case with  $m_i = 0$  in Table A.6(a). The main findings are as follows: (i) When  $m_i$  is the bump function, the test is conservative. In particular, the size numbers are substantially below the target  $\alpha$  for the smallest sample size  $T = 100$ , gradually getting better as the sample size increases. (ii) When  $m_i$  is the zero function, in contrast, the size numbers tend to be rather on the liberal than the conservative side. (iii) For smaller sample sizes, in particular for  $T = 100$ , the size numbers tend to be more accurate when  $m_i$  is the zero function. For larger sample sizes, in contrast, in particular

Figure A.2: Histograms of the estimated (square root of the) long-run error variance with  $m_i$  the zero and the bump function. The red vertical line is the true value of the (square root of the) long-run error variance.



for  $T = 500$ , the size numbers are fairly accurate no matter whether  $m_i$  is the zero or the bump function. How can these differences in size numbers be explained? Importantly, estimating the long-run error variance  $\sigma^2$  is more difficult in the presence of a non-trivial (i.e., non-constant) trend  $m_i$ . In particular, if we do not manage to eliminate the trend  $m_i$  accurately from the data (e.g. by differencing or nonparametric regression techniques), this will bias the estimator of the long-run error variance. This is illustrated in Figure A.2: whereas the estimates of the long-run error variance are nearly unbiased in the case with  $m_i = 0$  for all  $i$ , there is a clear bias in the case where  $m_i$  is the bump function (the bias getting smaller as the sample size increases). As the long-run error variance enters our multiscale test statistic  $\hat{\Psi}_{n,T}$  as a normalization constant, a biased estimate of it will in



general lead to size distortions.

### A.2.2 Performance of the test for large $n$

So far, we have fixed the number of time series at  $n = 15$ . We now examine the performance of our test as  $n$  gets larger. To do so, we re-run the simulation exercises from Section A.1 for  $n \in \{25, 50, 100\}$  and  $T$  as before, i.e.,  $T \in \{100, 250, 500\}$ . We work with precisely the same simulation setup as in Section A.1 except that we use the smaller grid  $\mathcal{G}_T^{\text{sparse}}$  defined in (A.4) to decrease the computational burden. Overall, our simulation results suggest that the test performs well for larger  $n$ . The results are presented in Table A.7, with each column corresponding to a particular choice of  $n$ . The upper panels (a) of Table A.7 show that the empirical size of the test provides a reasonably good approximation to the target  $\alpha$  across all scenarios under consideration. In particular, the size numbers remain fairly stable as  $n$  increases, even though they appear to get a little less precise for larger  $n$ . Panels (b)–(d) further show that the test has reasonable power across all considered scenarios. The power numbers, however, decrease somewhat as  $n$  gets larger, which may be explained as follows: Let  $\mathcal{M} = \{(u, h, i, j) : (u, h) \in \mathcal{G}_T \text{ and } 1 \leq i < j \leq n\}$  be the collection of all location-scale points  $(u, h)$  and all pairs of time series  $(i, j)$  under consideration and let  $\mathcal{M}_1 \subseteq \mathcal{M}$  be the set of tuples  $(u, h, i, j)$  for which the local null hypothesis  $H_0^{[i,j]}(u, h)$  is false. Since in the simulation setup at hand, all trend functions are the same except the first one ( $m_1$  being the bump function from (A.1) with bump size  $b$  and  $m_i = 0$  for all  $i \neq 1$ ), it holds that  $|\mathcal{M}_1|/|\mathcal{M}| \rightarrow 0$  quite fast as  $n \rightarrow \infty$  (where  $|\mathcal{M}|$  denotes the cardinality of the set  $\mathcal{M}$ ). Hence, as  $n$  increases, the total number of hypotheses to be tested grows much faster than the number of hypotheses which are violated. As a consequence, it gets harder to detect these violations, which is reflected in somewhat lower power numbers.

Table A.7: Empirical size and power of the multiscale test. Each column corresponds to a particular choice of  $n$ . Panels (a)–(d) of each column have the same structure as in Table A.1.

| $n = 15$              |       |       | $n = 25$              |     |       | $n = 50$              |       |     | $n = 100$             |       |       |
|-----------------------|-------|-------|-----------------------|-----|-------|-----------------------|-------|-----|-----------------------|-------|-------|
| (a) $b = 0$           |       |       | (a) $b = 0$           |     |       | (a) $b = 0$           |       |     | (a) $b = 0$           |       |       |
| nominal size $\alpha$ |       |       | nominal size $\alpha$ |     |       | nominal size $\alpha$ |       |     | nominal size $\alpha$ |       |       |
| $T$                   | 0.01  | 0.05  | 0.1                   | $T$ | 0.01  | 0.05                  | 0.1   | $T$ | 0.01                  | 0.05  | 0.1   |
| 100                   | 0.008 | 0.036 | 0.070                 | 100 | 0.005 | 0.032                 | 0.065 | 100 | 0.005                 | 0.034 | 0.066 |
| 250                   | 0.014 | 0.055 | 0.100                 | 250 | 0.012 | 0.054                 | 0.098 | 250 | 0.015                 | 0.061 | 0.110 |
| 500                   | 0.013 | 0.055 | 0.109                 | 500 | 0.018 | 0.059                 | 0.108 | 500 | 0.010                 | 0.068 | 0.128 |
| (b) $b = 0.25$        |       |       | (b) $b = 0.25$        |     |       | (b) $b = 0.25$        |       |     | (b) $b = 0.25$        |       |       |
| nominal size $\alpha$ |       |       | nominal size $\alpha$ |     |       | nominal size $\alpha$ |       |     | nominal size $\alpha$ |       |       |
| $T$                   | 0.01  | 0.05  | 0.1                   | $T$ | 0.01  | 0.05                  | 0.1   | $T$ | 0.01                  | 0.05  | 0.1   |
| 100                   | 0.008 | 0.042 | 0.082                 | 100 | 0.008 | 0.042                 | 0.080 | 100 | 0.008                 | 0.039 | 0.078 |
| 250                   | 0.078 | 0.174 | 0.261                 | 250 | 0.056 | 0.159                 | 0.237 | 250 | 0.046                 | 0.134 | 0.215 |
| 500                   | 0.203 | 0.438 | 0.576                 | 500 | 0.181 | 0.378                 | 0.512 | 500 | 0.106                 | 0.329 | 0.453 |
| (c) $b = 0.5$         |       |       | (c) $b = 0.5$         |     |       | (c) $b = 0.5$         |       |     | (c) $b = 0.5$         |       |       |
| nominal size $\alpha$ |       |       | nominal size $\alpha$ |     |       | nominal size $\alpha$ |       |     | nominal size $\alpha$ |       |       |
| $T$                   | 0.01  | 0.05  | 0.1                   | $T$ | 0.01  | 0.05                  | 0.1   | $T$ | 0.01                  | 0.05  | 0.1   |
| 100                   | 0.011 | 0.047 | 0.100                 | 100 | 0.008 | 0.045                 | 0.089 | 100 | 0.008                 | 0.041 | 0.077 |
| 250                   | 0.501 | 0.709 | 0.807                 | 250 | 0.410 | 0.634                 | 0.739 | 250 | 0.364                 | 0.589 | 0.691 |
| 500                   | 0.995 | 0.999 | 0.999                 | 500 | 0.994 | 0.999                 | 1.000 | 500 | 0.983                 | 0.999 | 1.000 |
| (d) $b = 0.75$        |       |       | (d) $b = 0.75$        |     |       | (d) $b = 0.75$        |       |     | (d) $b = 0.75$        |       |       |
| nominal size $\alpha$ |       |       | nominal size $\alpha$ |     |       | nominal size $\alpha$ |       |     | nominal size $\alpha$ |       |       |
| $T$                   | 0.01  | 0.05  | 0.1                   | $T$ | 0.01  | 0.05                  | 0.1   | $T$ | 0.01                  | 0.05  | 0.1   |
| 100                   | 0.007 | 0.027 | 0.064                 | 100 | 0.005 | 0.032                 | 0.062 | 100 | 0.006                 | 0.033 | 0.063 |
| 250                   | 0.743 | 0.893 | 0.946                 | 250 | 0.656 | 0.848                 | 0.907 | 250 | 0.615                 | 0.798 | 0.874 |
| 500                   | 1.000 | 1.000 | 1.000                 | 500 | 1.000 | 1.000                 | 1.000 | 500 | 1.000                 | 1.000 | 1.000 |

### A.2.3 Robustness checks on the choice of tuning parameters

The main tuning parameter of our method is the grid  $\mathcal{G}_T$ , which in particular specifies the bandwidths that are taken into account. In order to assess the robustness of our test with respect to the grid, we repeat the simulation exercises from Section A.1 with two different choices of the grid: a sparser and a finer grid than the one used in Section A.1. The sparser grid is defined as

$$\begin{aligned} \mathcal{G}_T^{\text{sparse}} = \left\{ (u, h) \subseteq [0, 1] : (u, h) = ((2s+1)h, h) \text{ for } s = 0, \dots, \left\lfloor \frac{h^{-1} - 1}{2} \right\rfloor \right. \\ \left. \text{and } h \in H_T^{\text{sparse}} \right\} \end{aligned} \quad (\text{A.4})$$

with  $H_T^{\text{sparse}} = \{h = 2^k h_{\min} \text{ for } k = 0, \dots, K\}$ , where  $h_{\min} = \frac{\lceil \log T \rceil}{T}$  and  $K$  is such that  $2^K h_{\min} \leq \frac{1}{4}$ , i.e.,  $K \leq \lfloor \log(\frac{T}{4 \lceil \log T \rceil}) \frac{1}{\log(2)} \rfloor$ . This is a dyadic scheme with scales in the set  $H_T^{\text{sparse}}$  as commonly used in Wavelet analysis. The finer grid is defined as  $\mathcal{G}_T^{\text{fine}} = U_T^{\text{fine}} \times H_T^{\text{fine}}$ , where  $U_T^{\text{fine}} = \{u \in [0, 1] : u = \frac{t}{T} \text{ with } 1 \leq t \leq T\}$  and  $H_T^{\text{fine}} = \{h \in [\frac{\log T}{T}, \frac{1}{4}] : h = \frac{2t}{T} \text{ with } 1 \leq t \leq T\}$ . Apart from the choice of grid, we work with precisely the same simulation setup as in Section A.1.<sup>1</sup>

The simulation results for the three grids are reported in Table A.8.<sup>2</sup> Each column corresponds to a particular choice of grid. Panel (a) in each column specifies the empirical size of the test, while panels (b)–(d) give the empirical power against the alternatives with bump size  $b \in \{0.25, 0.5, 0.75\}$ . As can be seen, the size numbers in Table A.8 are fairly stable across the three different grids. In particular, we get a reasonable approximation to the target size  $\alpha$ , no matter which of the three grids we are working with. Moreover, the power numbers for  $\mathcal{G}_T$  and the finer grid  $\mathcal{G}_T^{\text{fine}}$  are quite similar to each other. Those for the sparser grid  $\mathcal{G}_T^{\text{sparse}}$ , in contrast, are somewhat lower, which may be explained as

<sup>1</sup>As the grid  $\mathcal{G}_T^{\text{fine}}$  is very fine, the number of tests to be carried out is extremely large, which significantly increases the computational burden. To decrease computational time, we proceed as follows when using the fine grid  $\mathcal{G}_T^{\text{fine}}$ : we calculate the Gaussian quantile based on  $L = 1000$  (rather than 5000) bootstrap samples and we simulate 1000 samples (rather than 5000) for each simulation scenario.

<sup>2</sup>Note that the simulation results for the grid  $\mathcal{G}_T$  in the second column of Table A.8 are identical with those in Table A.1. For ease of comparison, we have repeated these results as part of Table A.8.

Table A.8: Empirical size and power of the multiscale test for three different grids. Each column corresponds to a particular choice of grid. Panels (a)–(d) of each column have the same structure as in Table A.1.

| sparser grid $\mathcal{G}_T^{\text{sparse}}$ |       |       |       | grid $\mathcal{G}_T$ from Section A.1 |       |       |       | finer grid $\mathcal{G}_T^{\text{fine}}$ |       |       |       |
|----------------------------------------------|-------|-------|-------|---------------------------------------|-------|-------|-------|------------------------------------------|-------|-------|-------|
| (a) $b = 0$                                  |       |       |       | (a) $b = 0$                           |       |       |       | (a) $b = 0$                              |       |       |       |
| nominal size $\alpha$                        |       |       |       | nominal size $\alpha$                 |       |       |       | nominal size $\alpha$                    |       |       |       |
| $T$                                          | 0.01  | 0.05  | 0.1   | $T$                                   | 0.01  | 0.05  | 0.1   | $T$                                      | 0.01  | 0.05  | 0.1   |
| 100                                          | 0.008 | 0.036 | 0.070 | 100                                   | 0.006 | 0.035 | 0.070 | 100                                      | 0.010 | 0.031 | 0.054 |
| 250                                          | 0.014 | 0.055 | 0.100 | 250                                   | 0.020 | 0.079 | 0.134 | 250                                      | 0.014 | 0.050 | 0.106 |
| 500                                          | 0.013 | 0.055 | 0.109 | 500                                   | 0.012 | 0.058 | 0.112 | 500                                      | 0.012 | 0.054 | 0.117 |
| (b) $b = 0.25$                               |       |       |       | (b) $b = 0.25$                        |       |       |       | (b) $b = 0.25$                           |       |       |       |
| nominal size $\alpha$                        |       |       |       | nominal size $\alpha$                 |       |       |       | nominal size $\alpha$                    |       |       |       |
| $T$                                          | 0.01  | 0.05  | 0.1   | $T$                                   | 0.01  | 0.05  | 0.1   | $T$                                      | 0.01  | 0.05  | 0.1   |
| 100                                          | 0.008 | 0.042 | 0.082 | 100                                   | 0.011 | 0.046 | 0.095 | 100                                      | 0.014 | 0.044 | 0.075 |
| 250                                          | 0.078 | 0.174 | 0.261 | 250                                   | 0.158 | 0.348 | 0.473 | 250                                      | 0.129 | 0.283 | 0.389 |
| 500                                          | 0.203 | 0.438 | 0.576 | 500                                   | 0.524 | 0.762 | 0.854 | 500                                      | 0.526 | 0.761 | 0.824 |
| (c) $b = 0.5$                                |       |       |       | (c) $b = 0.5$                         |       |       |       | (c) $b = 0.5$                            |       |       |       |
| nominal size $\alpha$                        |       |       |       | nominal size $\alpha$                 |       |       |       | nominal size $\alpha$                    |       |       |       |
| $T$                                          | 0.01  | 0.05  | 0.1   | $T$                                   | 0.01  | 0.05  | 0.1   | $T$                                      | 0.01  | 0.05  | 0.1   |
| 100                                          | 0.011 | 0.047 | 0.100 | 100                                   | 0.012 | 0.058 | 0.105 | 100                                      | 0.019 | 0.053 | 0.091 |
| 250                                          | 0.501 | 0.709 | 0.807 | 250                                   | 0.837 | 0.952 | 0.979 | 250                                      | 0.822 | 0.935 | 0.965 |
| 500                                          | 0.995 | 0.999 | 0.999 | 500                                   | 1.000 | 1.000 | 1.000 | 500                                      | 1.000 | 1.000 | 1.000 |
| (d) $b = 0.75$                               |       |       |       | (d) $b = 0.75$                        |       |       |       | (d) $b = 0.75$                           |       |       |       |
| nominal size $\alpha$                        |       |       |       | nominal size $\alpha$                 |       |       |       | nominal size $\alpha$                    |       |       |       |
| $T$                                          | 0.01  | 0.05  | 0.1   | $T$                                   | 0.01  | 0.05  | 0.1   | $T$                                      | 0.01  | 0.05  | 0.1   |
| 100                                          | 0.007 | 0.027 | 0.064 | 100                                   | 0.010 | 0.035 | 0.071 | 100                                      | 0.009 | 0.031 | 0.052 |
| 250                                          | 0.743 | 0.893 | 0.946 | 250                                   | 0.967 | 0.992 | 0.996 | 250                                      | 0.967 | 0.992 | 0.997 |
| 500                                          | 1.000 | 1.000 | 1.000 | 500                                   | 1.000 | 1.000 | 1.000 | 500                                      | 1.000 | 1.000 | 1.000 |

follows: As  $\mathcal{G}_T^{\text{sparse}}$  is considerably sparser than the other two grids, much less intervals  $\mathcal{I}_{u,h} = [u - h, u + h]$  are taken into account. In particular, intervals may potentially be ignored where the deviation from the null is quite strong. This results in a loss of power. Taken together, our simulation results suggest the following: (i) The particular choice of grid has a negligible effect on the empirical size of the test. (ii) As long as the grid is sufficiently rich, the choice of grid also has a negligible effect on the power of the test.

Apart from the grid, our method depends on the following tuning parameters: (i) the num-

ber of bootstrap samples  $L$  to compute the Gaussian quantile, (ii) the kernel  $K$ , and (iii) secondary tuning parameters for the computation of the long-run error variance. As long as  $L$  is chosen large enough (say  $L \geq 1000$ ), the exact value of  $L$  should have a negligible effect. We use  $L = 5000$  throughout the paper.<sup>3</sup> As a robustness check, we have rerun everything with  $L = 1000$ , which (as expected) yields almost identical results.<sup>4</sup> As suggested by classical nonparametric theory, the Epanechnikov kernel (which is used throughout the paper) has good properties and the choice of kernel  $K$  is of minor importance, in particular, much less important than the choice of bandwidth. Thus, we have not carried out robustness checks with respect to the choice of kernel. Our procedure can be combined with any method off-the-shelf for long-run error variance estimation. In the current simulation study, we work with the long-run variance estimator from Khismatullina and Vogt (2020), where extensive robustness checks with respect to the choice of tuning parameters have been carried out. We refer to Section 5.2 therein for the details.

#### A.2.4 Performance of the test in the presence of time-varying error variance

In what follows, we investigate the performance of our test in the presence of time-varying error variance. We consider the same simulation setup as in Section A.1 except that we model the error processes  $\{\varepsilon_{it} : 1 \leq t \leq T\}$  slightly differently. Specifically, for each  $i$ , we consider an AR(1) process of the form  $\varepsilon_{it} = a\varepsilon_{it-1} + \nu(t/T)\eta_{it}$  with  $a = 0.25$  and i.i.d. standard normal innovations  $\eta_{it}$ . In the simulation setup of Section A.1, the function  $\nu$  was assumed to be constant, in particular,  $\nu(u) = c_\nu$  with  $c_\nu = 0.25$  for all  $u \in [0, 1]$ . Now, in contrast, we allow  $\nu$  to vary over time, specifically, we let  $\nu(u) = \{-0.15(u-0.5)^2 + 0.075\}^{1/2}$  for  $u \in [0, 1]$ . Note that with this choice, the average short-run error variance  $\int_0^1 \nu^2(u)du$  is equal to  $c_\nu^2 = 0.25^2$  and thus matches the short-run error variance in the setup of Section A.1. Notably, the time variation induced by our choice of  $\nu$  is quite strong. In particular, we

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<sup>3</sup>Except for the simulations with the fine grid  $\mathcal{G}_T^{\text{fine}}$ , where  $L = 1000$ .

<sup>4</sup>The results are not reported here but are available from the authors upon request.

Table A.9: Empirical size and power of the multiscale test in case of time-varying error variance for different time series lengths  $T$  and nominal sizes  $\alpha$ . Each panel gives the results for a different bump height  $b$ , where  $b = 0$  corresponds to the null  $H_0$  and  $b = 0.25, 0.5, 0.75$  to three different alternatives.

| (a) $b = 0$           |       |       |       | (b) $b = 0.25$        |       |       |       |
|-----------------------|-------|-------|-------|-----------------------|-------|-------|-------|
| nominal size $\alpha$ |       |       |       | nominal size $\alpha$ |       |       |       |
| $T$                   | 0.01  | 0.05  | 0.1   | $T$                   | 0.01  | 0.05  | 0.1   |
| 100                   | 0.001 | 0.008 | 0.018 | 100                   | 0.014 | 0.072 | 0.135 |
| 250                   | 0.007 | 0.021 | 0.038 | 250                   | 0.212 | 0.411 | 0.543 |
| 500                   | 0.007 | 0.020 | 0.034 | 500                   | 0.622 | 0.821 | 0.890 |
| (c) $b = 0.5$         |       |       |       | (d) $b = 0.75$        |       |       |       |
| nominal size $\alpha$ |       |       |       | nominal size $\alpha$ |       |       |       |
| $T$                   | 0.01  | 0.05  | 0.1   | $T$                   | 0.01  | 0.05  | 0.1   |
| 100                   | 0.019 | 0.077 | 0.140 | 100                   | 0.009 | 0.052 | 0.098 |
| 250                   | 0.876 | 0.962 | 0.984 | 250                   | 0.976 | 0.994 | 0.997 |
| 500                   | 1.000 | 1.000 | 1.000 | 500                   | 1.000 | 1.000 | 1.000 |

have  $\max_{u \in [0,1]} \nu^2(u) = 2 \min_{u \in [0,1]} \nu^2(u)$ . The simulation results are presented in Table A.9. Even though the test has reasonable power properties (panels (b)–(d) in Table A.9), the empirical size lies substantially below the target  $\alpha$  (panel (a) in Table A.9), implying that the test is quite conservative in the case at hand. This demonstrates that the test should be treated with caution when the (short-run and/or long-run) error variance is expected to be strongly time-varying. The issue is that the scaling of our test statistics depends on the long-run error variance (see equation (3.6)) and that this scaling is theoretically not correct any more if the long-run error variance is time-varying. We may therefore expect non-negligible size distortions when the long-run error variance strongly varies over time.

### A.3 Comparison of our test with competing methods

We compare our methods with two competitors. First, we consider the following simple procedure: (i) compute a uniform confidence band (UCB) at level  $(1 - \alpha)$  for each trend function  $m_i$  and (ii) reject the hypothesis that  $m_i$  and  $m_j$  are the same if their UCBs do not overlap at some time point. This procedure may be run with a Bonferroni correction

to take into account that multiple comparisons are carried out. In addition, we compare our test method with the SiZer test from Park et al. (2009).

### A.3.1 Comparison with a simple competitor

To set up the first competitor, we compute uniform confidence bands (UCBs) for the trends  $m_i$  as described in Kim (2016). For our comparison, we consider the simplified model  $Y_{it} = m_i(\frac{t}{T}) + \beta_i^\top \mathbf{X}_{it} + \varepsilon_{it}$  without fixed effects since the specification in Kim (2016) does not allow for them. Other than that, the simulation setup is as specified in Section A.1. In particular, to generate data under the null  $H_0 : m_1 = \dots = m_n$ , we let  $m_i = 0$  for all  $i$ , and to produce data under the alternative, we use the bump function  $m_1$  defined in (A.1) with bump size  $b \in \{0.25, 0.5, 0.75\}$  while setting  $m_i = 0$  for  $i \neq 1$ . We implement our multiscale test as in Section A.1, where we fix the nominal size at  $\alpha = 0.05$ . To implement the UCBs, we set  $\alpha = 0.05$  and estimate the trend function  $m_i$  separately for each time series  $i$  by a local linear estimate with bandwidth  $h = 0.2$ .<sup>5</sup> We then construct the UCB of  $m_i$  as described in Kim (2016). In what follows, we denote our multiscale test by  $\mathcal{T}_{\text{MS}}$  and the test based on the UCBs from Kim (2016) by  $\mathcal{T}_{\text{UCB}}$ .

Table A.10 compares the size and power properties of the two tests: panel (a) reports the empirical size numbers, while panels (b)–(d) specify the empirical power numbers under the three different alternatives with bump size  $b \in \{0.25, 0.5, 0.75\}$ . To calculate the empirical size and power for  $\mathcal{T}_{\text{UCB}}$ , we consider the null hypothesis  $H_0$  as rejected if there exists at least one pair  $(i, j)$  and at least one time point  $t/T$  such that the UCBs of  $m_i$  and  $m_j$  do not overlap at  $t/T$ . As the estimation procedure from Kim (2016) turned out to perform extremely poorly at the boundaries, we restrict attention to points  $t/T \in [0.05, 0.95]$  (which is fine as in our simulation setting, the trend  $m_1$  differs from the other trends only in the

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<sup>5</sup>Kim (2016) proposes a data-driven way to calculate the bandwidth using cross-validation, which needs to be repeated for each realisation of the data and is therefore computationally very expensive. Hence, we have chosen several fixed bandwidths ( $h = 0.1, 0.2, 0.3$ ) and report here the results for the best-performing one (which is  $h = 0.2$ ).

Table A.10: Size and power comparisons (with  $\alpha$  fixed at 0.05) for our multiscale test  $\mathcal{T}_{\text{MS}}$  and the procedure  $\mathcal{T}_{\text{UCB}}$  based on UCBs from Kim (2016).

| (a) $b = 0$ |                           |                            | (b) $b = 0.25$ |                           |                            | (c) $b = 0.5$ |                           |                            | (d) $b = 0.75$ |                           |                            |
|-------------|---------------------------|----------------------------|----------------|---------------------------|----------------------------|---------------|---------------------------|----------------------------|----------------|---------------------------|----------------------------|
| $T$         | $\mathcal{T}_{\text{MS}}$ | $\mathcal{T}_{\text{UCB}}$ | $T$            | $\mathcal{T}_{\text{MS}}$ | $\mathcal{T}_{\text{UCB}}$ | $T$           | $\mathcal{T}_{\text{MS}}$ | $\mathcal{T}_{\text{UCB}}$ | $T$            | $\mathcal{T}_{\text{MS}}$ | $\mathcal{T}_{\text{UCB}}$ |
| 100         | 0.036                     | 0.000                      | 100            | 0.045                     | 0.000                      | 100           | 0.039                     | 0.002                      | 100            | 0.029                     | 0.010                      |
| 250         | 0.087                     | 0.000                      | 250            | 0.350                     | 0.001                      | 250           | 0.962                     | 0.029                      | 250            | 0.995                     | 0.638                      |
| 500         | 0.067                     | 0.000                      | 500            | 0.754                     | 0.000                      | 500           | 1.000                     | 0.654                      | 500            | 1.000                     | 1.000                      |

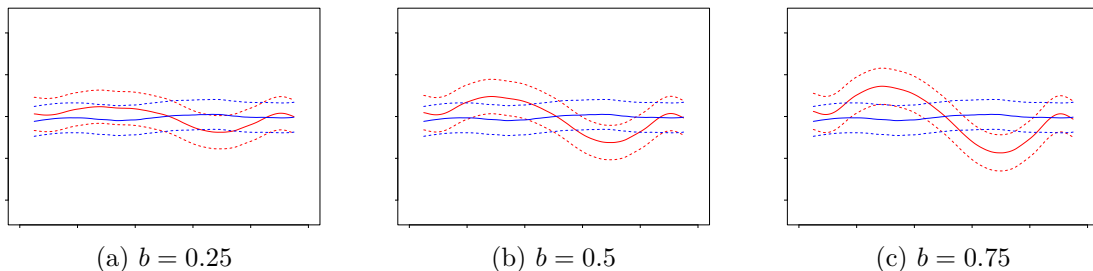


Figure A.3: Estimated trends  $m_1$  (solid line in red) and  $m_2 = 0$  (solid line in blue) and their corresponding UCBs (dashed lines) for  $T = 500$  and  $b = 0.25, 0.5, 0.75$ .

time period  $\mathcal{I}^* = [0.2, 0.4] \cup [0.6, 0.8]$  anyway). Table A.10 demonstrates that our test has reasonable size and power properties. The test  $\mathcal{T}_{\text{UCB}}$ , in contrast, is extremely conservative (with empirical size exactly equal to 0 for all considered levels  $\alpha$ !) and has extremely low power for smaller sample sizes, the power moderately increasing with the sample size. Notably, we have run  $\mathcal{T}_{\text{UCB}}$  without any Bonferroni correction which would make things even worse! The reason for this poor performance is illustrated in Figure A.3 which shows the estimated trends  $m_1$  and  $m_2$  together with their UCBs in a representative simulation run for the largest sample size  $T = 500$ . Even for this sample size, the bands are still very wide, overlapping everywhere for the bump sizes  $b = 0.25$  and  $b = 0.5$ . Only for the largest bump size  $b = 0.75$ , the bands do not overlap at all time points (but only hardly so). Hence, the computed UCBs are much too wide to produce a good test performance.

### A.3.2 Comparison with SiZer

To the best of our knowledge, the only other multiscale test for comparing trend curves available in the literature is the SiZer test developed in Park et al. (2009). In this section,



we compare our method with this test. Notably, Park et al. (2009) consider a simplified version of our model without covariates and fixed effects. Moreover, rigorous theory has been developed only for the case of  $n = 2$  time series. The authors propose a possible extension of their approach to the case of more than  $n = 2$  time series, but the extension does not allow for a pairwise comparison of the time series. Moreover, it is not clear how to calculate the actual size and power of the test in the case of more than  $n = 2$  time series. For our comparison, we thus consider the model  $Y_{it} = m_i\left(\frac{t}{T}\right) + \varepsilon_{it}$  without covariates and fixed effects and we choose the number of time series to be equal to  $n = 2$ .

The simulation setup is as specified in Section A.1 (without covariates and fixed effects, i.e.,  $\beta_i = 0$  and  $\alpha_i = 0$  for all  $i$ ). As before, to generate data under the null  $H_0 : m_1 = m_2$ , we let  $m_i = 0$  for  $i = 1, 2$ , and to produce data under the alternative, we use the bump function  $m_1$  defined in (A.1) with bump size  $b \in \{0.25, 0.5\}$  while setting  $m_2 = 0$ . We implement our multiscale test as in Section A.1. For the implementation of the SiZer test, we use the same location-scale grid  $\mathcal{G}_T$  as for our test. While our method depends on the long-run error variance  $\sigma^2$ , the SiZer test depends on the whole autocovariance function  $\{\gamma(k) : k \in \mathbb{Z}\}$  of the error process. To make the comparison as fair as possible, we assume the autocovariance function  $\gamma(\cdot)$  and thus the long-run error variance  $\sigma^2 = \sum_{k=-\infty}^{\infty} \gamma(k)$  to be known when implementing the two methods.

As before, we use the notation  $\mathcal{T}_{\text{MS}}$  for our multiscale test and we denote the SiZer test of Park et al. (2009) by  $\mathcal{T}_{\text{SiZer}}$ . Table A.11 presents the results of our comparison study. Panel (a) compares the empirical size of the two tests, while panels (b) and (c) provide a power comparison. There is an important difference between  $\mathcal{T}_{\text{MS}}$  and  $\mathcal{T}_{\text{SiZer}}$  to be noticed:  $\mathcal{T}_{\text{MS}}$  is a global test procedure, whereas  $\mathcal{T}_{\text{SiZer}}$  is scale-wise in essence. This means that  $\mathcal{T}_{\text{MS}}$  tests the hypotheses  $H_0^{[1,2]}(u, h)$  simultaneously for all locations  $u \in U_T$  and scales  $h \in H_T$ . In particular, it controls the size simultaneously over both locations  $u$  and scales  $h$ .  $\mathcal{T}_{\text{SiZer}}$ , in contrast, tests  $H_0^{[1,2]}(u, h)$  simultaneously for all  $u \in U_T$  but separately for each scale

Table A.11: Size and power comparison of our multiscale test  $\mathcal{T}_{\text{MS}}$  and the SiZer test  $\mathcal{T}_{\text{SiZer}}$  from Park et al. (2009).

| (a) $b = 0$ |                           |                              |                           |                              |                           |                              |
|-------------|---------------------------|------------------------------|---------------------------|------------------------------|---------------------------|------------------------------|
|             | $\alpha = 0.01$           |                              | $\alpha = 0.05$           |                              | $\alpha = 0.1$            |                              |
|             | $\mathcal{T}_{\text{MS}}$ | $\mathcal{T}_{\text{SiZer}}$ | $\mathcal{T}_{\text{MS}}$ | $\mathcal{T}_{\text{SiZer}}$ | $\mathcal{T}_{\text{MS}}$ | $\mathcal{T}_{\text{SiZer}}$ |
| $T = 100$   | 0.010                     | 0.099                        | 0.047                     | 0.332                        | 0.106                     | 0.524                        |
| $T = 250$   | 0.008                     | 0.162                        | 0.041                     | 0.510                        | 0.102                     | 0.699                        |
| $T = 500$   | 0.009                     | 0.214                        | 0.044                     | 0.598                        | 0.089                     | 0.787                        |

| (b) $b = 0.25$ |                           |                              |                           |                              |                           |                              |
|----------------|---------------------------|------------------------------|---------------------------|------------------------------|---------------------------|------------------------------|
|                | $\alpha = 0.01$           |                              | $\alpha = 0.05$           |                              | $\alpha = 0.1$            |                              |
|                | $\mathcal{T}_{\text{MS}}$ | $\mathcal{T}_{\text{SiZer}}$ | $\mathcal{T}_{\text{MS}}$ | $\mathcal{T}_{\text{SiZer}}$ | $\mathcal{T}_{\text{MS}}$ | $\mathcal{T}_{\text{SiZer}}$ |
| $T = 100$      | 0.075                     | 0.251                        | 0.193                     | 0.551                        | 0.317                     | 0.714                        |
| $T = 250$      | 0.253                     | 0.629                        | 0.486                     | 0.876                        | 0.633                     | 0.951                        |
| $T = 500$      | 0.640                     | 0.905                        | 0.817                     | 0.985                        | 0.887                     | 0.996                        |

| (c) $b = 0.5$ |                           |                              |                           |                              |                           |                              |
|---------------|---------------------------|------------------------------|---------------------------|------------------------------|---------------------------|------------------------------|
|               | $\alpha = 0.01$           |                              | $\alpha = 0.05$           |                              | $\alpha = 0.1$            |                              |
|               | $\mathcal{T}_{\text{MS}}$ | $\mathcal{T}_{\text{SiZer}}$ | $\mathcal{T}_{\text{MS}}$ | $\mathcal{T}_{\text{SiZer}}$ | $\mathcal{T}_{\text{MS}}$ | $\mathcal{T}_{\text{SiZer}}$ |
| $T = 100$     | 0.482                     | 0.733                        | 0.703                     | 0.922                        | 0.815                     | 0.967                        |
| $T = 250$     | 0.966                     | 0.998                        | 0.994                     | 1.000                        | 0.998                     | 1.000                        |
| $T = 500$     | 1.000                     | 1.000                        | 1.000                     | 1.000                        | 1.000                     | 1.000                        |

$h \in H_T$  and, hence, controls the size for each scale  $h \in H_T$  separately. This results in  $\mathcal{T}_{\text{SiZer}}$  being much too liberal, as panel (a) illustrates: whereas the empirical size numbers of  $\mathcal{T}_{\text{MS}}$  closely align with the target size  $\alpha$ , the size numbers of  $\mathcal{T}_{\text{SiZer}}$  are much larger than the target  $\alpha$ . In particular, they deviate more and more strongly from the target  $\alpha$  as  $T$  increases, since the number of scales  $h$  in the grid  $\mathcal{G}_T$  increases with  $T$ . To summarize, as expected, the global test  $\mathcal{T}_{\text{MS}}$  holds the size reasonably well, whereas the scale-wise method  $\mathcal{T}_{\text{SiZer}}$  is much too liberal. Inspecting the power numbers in panels (b) and (c), one can further see that in all simulation scenarios,  $\mathcal{T}_{\text{SiZer}}$  is more powerful than  $\mathcal{T}_{\text{MS}}$ . These higher power numbers, however, should not be taken at face value. In particular, they do not imply that  $\mathcal{T}_{\text{SiZer}}$  is better in detecting deviations from the null than our test. They are rather a simple consequence of the fact that  $\mathcal{T}_{\text{SiZer}}$  is much too liberal as illustrated in panel (a).

## A.4 Performance of the clustering algorithm

We next investigate the finite sample performance of the clustering algorithm from Section 5. To do so, we consider the same simulation setup as in Section A.1 with one exception: we partition the  $n = 15$  time series into  $N = 3$  groups, each containing 5 time series. Specifically, we set  $G_1 = \{1, \dots, 5\}$ ,  $G_2 = \{6, \dots, 10\}$  and  $G_3 = \{11, \dots, 15\}$ , and we assume that  $m_i = f_l$  for all  $i \in G_l$  and all  $l = 1, 2, 3$ . The group-specific trend functions  $f_1$ ,  $f_2$  and  $f_3$  are defined as  $f_1(u) = 0$ ,  $f_2(u) = 0.35 \cdot \vartheta(u, 0.25, 0.25) - 0.35 \cdot \vartheta(u, 0.75, 0.25)$  and  $f_3(u) = \vartheta(u, 0.75, 0.025) - \vartheta(u, 0.25, 0.025)$  with  $\vartheta(u, c, d) = \mathbb{1}\left\{\frac{|u-c|}{d} \leq 1\right\} \left(1 - \frac{(u-c)^2}{d^2}\right)^2$  (depicted in Figure A.4). In order to estimate the groups  $G_1$ ,  $G_2$ ,  $G_3$  and their number  $N = 3$ , we use the same implementation as in Section A.1 followed by the clustering procedure from Section 5.

The simulation results are reported in Table A.12. The entries in Table A.12(a) are computed as the number of simulations for which  $\hat{N} = N$  divided by the total number of simulations. They thus specify the empirical probabilities with which the estimate  $\hat{N}$  is equal to the true number of groups  $N = 3$ . Analogously, the entries of Table A.12(b) give the empirical probabilities with which the estimated group structure  $\{\hat{G}_1, \dots, \hat{G}_{\hat{N}}\}$  equals the true one  $\{G_1, G_2, G_3\}$ . The results in Table A.12 nicely illustrate the theoretical properties of our clustering algorithm. According to Proposition 5.1, the probability that  $\hat{N} = N$  and that  $\{\hat{G}_1, \dots, \hat{G}_{\hat{N}}\} = \{G_1, G_2, G_3\}$  should be at least  $(1 - \alpha)$  asymptotically. For the largest sample size  $T = 500$ , the empirical probabilities reported in Table A.12 can indeed be seen to (almost) exceed the value  $(1 - \alpha)$  for  $\alpha \in \{0.05, 0.1\}$  as predicted by Proposition 5.1. For the smaller sample sizes  $T = 100$  and  $T = 250$ , in contrast, the empirical probabilities are substantially smaller than  $(1 - \alpha)$ . This reflects the asymptotic nature of Proposition 5.1 and is not very surprising. It simply mirrors the fact that for the smaller sample sizes  $T = 100$  and  $T = 250$ , the noise level in the simulated data is very

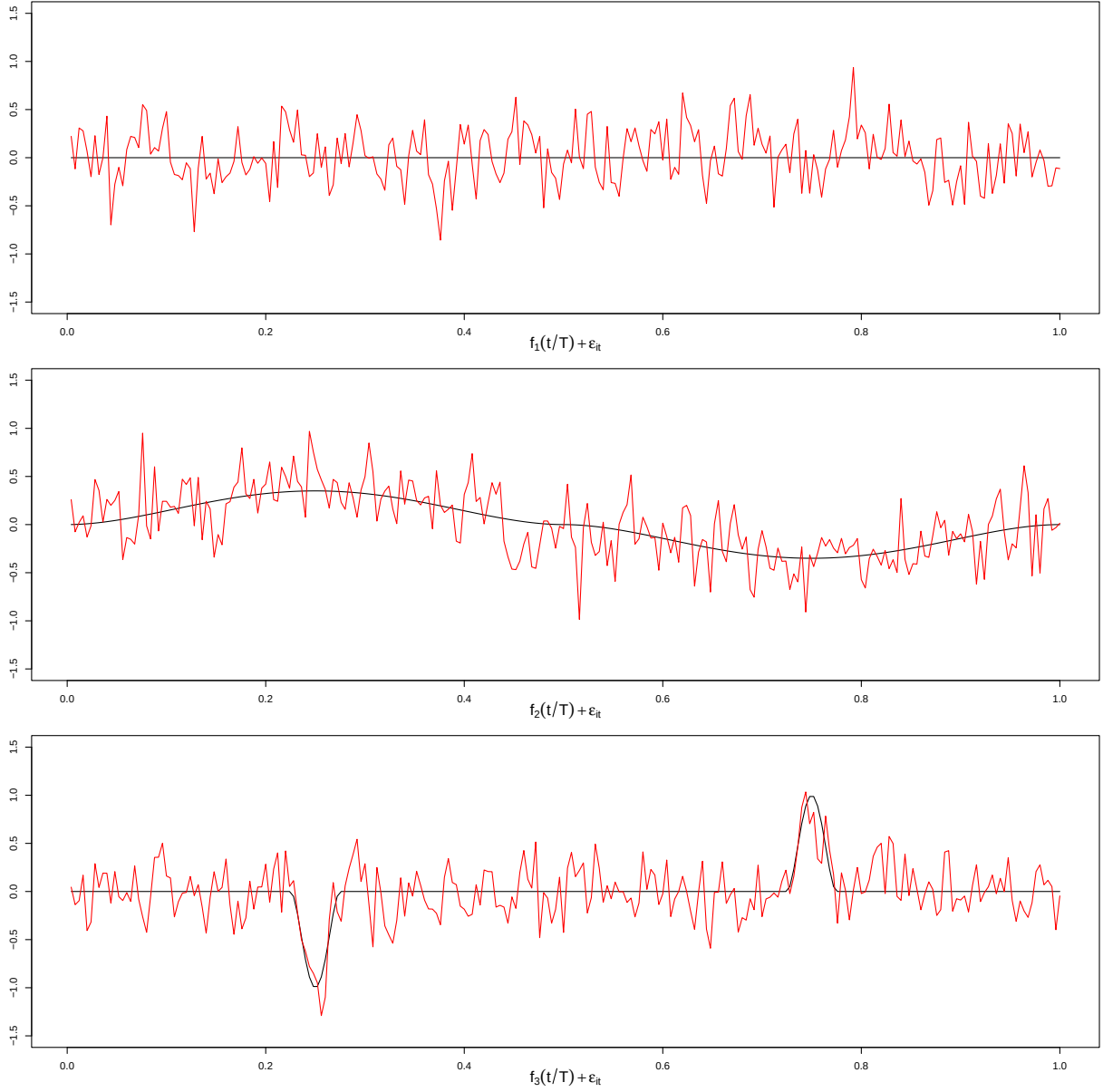


Figure A.4: In black: the group-specific trend functions  $f_l$  ( $l = 1, 2, 3$ ). In red: the time series  $\{Y_{it}^\circ : 1 \leq t \leq T\}$  with  $Y_{it}^\circ = f_l(t/T) + \varepsilon_{it}$  ( $l = 1, 2, 3$ ) for one simulation run and  $T = 250$ .

high.

Figures A.5 and A.6 give more insight into what happens for  $T = 100$  and  $T = 250$ .

Figure A.5 shows histograms of the 5000 simulated values of  $\hat{N}$ , while Figure A.6 depicts histograms of the number of classification errors produced by our algorithm. By the number of classification errors, we simply mean the number of incorrectly classified time series, which is formally calculated as  $\min_{\pi \in \Pi} \{|G_1 \setminus \hat{G}_{\pi(1)}| + \dots + |G_N \setminus \hat{G}_{\pi(N)}|\}$ , where  $\Pi$  is the

Table A.12: Clustering results for different time series lengths  $T$  and nominal sizes  $\alpha$ .

(a) Empirical probabilities that  $\hat{N} = N$

| $T$ | nominal size $\alpha$ |       |       |
|-----|-----------------------|-------|-------|
|     | 0.01                  | 0.05  | 0.1   |
| 100 | 0.000                 | 0.002 | 0.005 |
| 250 | 0.004                 | 0.023 | 0.051 |
| 500 | 0.758                 | 0.953 | 0.964 |

(b) Empirical probabilities that  $\{\hat{G}_1, \dots, \hat{G}_{\hat{N}}\} = \{G_1, G_2, G_3\}$

| $T$ | nominal size $\alpha$ |       |       |
|-----|-----------------------|-------|-------|
|     | 0.01                  | 0.05  | 0.1   |
| 100 | 0.000                 | 0.000 | 0.000 |
| 250 | 0.001                 | 0.008 | 0.020 |
| 500 | 0.751                 | 0.947 | 0.958 |

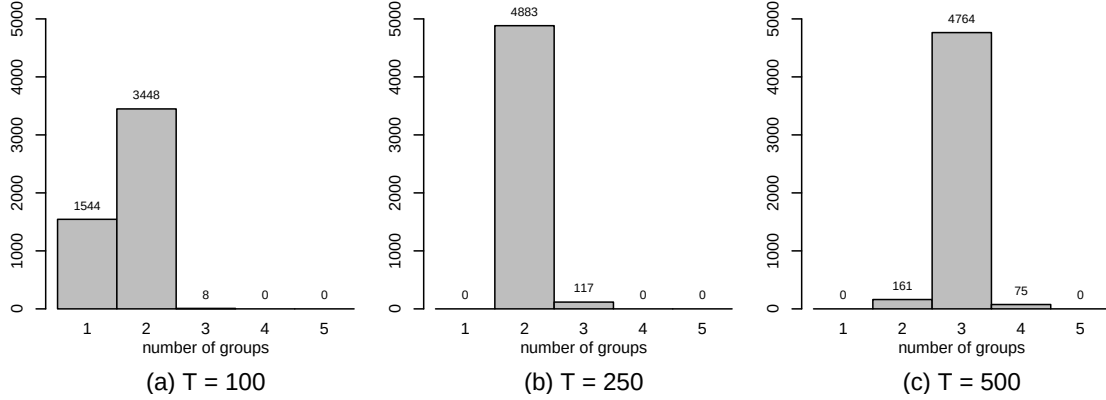


Figure A.5: Histograms of the estimated number of groups  $\hat{N}$  for nominal size  $\alpha = 0.05$ .

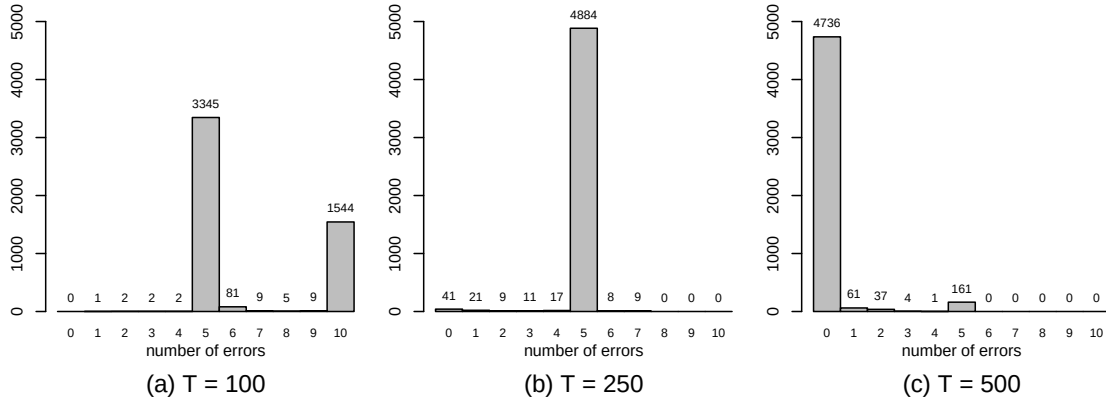


Figure A.6: Histograms of the number of classification errors for nominal size  $\alpha = 0.05$ .

set of all permutations of  $\{1, 2, \dots, \max\{N, \hat{N}\}\}$  and we let  $\hat{G}_l = \emptyset$  for  $\hat{N} < l \leq N$ . The histograms of Figure A.5 for  $T = 100$  and  $T = 250$  clearly show that our method underestimates the number of groups. In particular, it fails to detect the difference between two out of three groups in a large number of simulations ( $\hat{N} = 2$  in 3448 and 4883 cases out of 5000 for  $T = 100$  and  $T = 250$ , respectively). This is reflected in the corresponding

histograms of Figure A.6 which show that there are exactly 5 classification errors in 3345 (4884) of the 5000 simulation runs for  $T = 100$  ( $T = 250$ ). In most of these cases, the estimated group structure  $\{\widehat{G}_1, \widehat{G}_2\}$  coincides with either  $\{G_1 \cup G_2, G_3\}$ ,  $\{G_1, G_2 \cup G_3\}$  or  $\{G_1 \cup G_3, G_2\}$ . In summary, we can conclude that the very small empirical probabilities for  $T = 100$  and  $T = 250$  in Table A.12 are due to the algorithm underestimating the number of groups. Inspecting the histograms for  $T = 500$  in Figures A.5 and A.6, one can see that the performance of the estimators  $\widehat{N}$  and  $\{\widehat{G}_1, \dots, \widehat{G}_{\widehat{N}}\}$  improves significantly as the time series length  $T$  increases.

## A.5 Comparison of our clustering algorithm with a benchmark

We finally compare our clustering algorithm with a simple benchmark. To do so, we consider a simplified version of the simulation setup described in Section A.4. In particular, we use the same setup but without covariates and fixed effects (i.e.,  $\beta_i = 0$  and  $\alpha_i = 0$  for all  $i$ ). The benchmark method proceeds as follows:

*Step 1.* For each  $i$ , estimate the trend function  $m_i$  by a standard local linear estimator

$$\widehat{m}_i = \widehat{m}_{i,h} \text{ with bandwidth } h.$$

*Step 2.* Measure the distance between  $\widehat{m}_i$  and  $\widehat{m}_j$  by the supremum metric

$$d_{ij}^{\text{bmk}} = \sup_{u \in [0,1]} |\widehat{m}_i(u) - \widehat{m}_j(u)|.$$

*Step 3.* Construct the following dissimilarity measure from these supremum distances:

$$\widehat{\Delta}^{\text{bmk}}(S, S') = \max_{i \in S, j \in S'} d_{ij}^{\text{bmk}}.$$

*Step 4.* Run a HAC algorithm with the computed dissimilarities.

Our multiscale approach can be regarded as a refinement of this very natural and intuitive benchmark method: it replaces the simple bandwidth-dependent distance measure  $d_{ij}^{\text{bmk}}$  by the multiscale distance measure  $d_{ij} = \max_{(u,h) \in \mathcal{G}_T} \widehat{\psi}_{ij,T}^0(u, h)$  and provides a way to estimate the number of clusters  $N$ . As the benchmark algorithm does not come with an estimate of

Table A.13: Empirical probabilities that the estimated group structure is identical to the true group structure  $\{G_1, G_2, G_3\}$ .  $\mathcal{T}_{\text{MS}}$  denotes our multiscale method and  $\mathcal{T}_{\text{bmk}, l}$  the benchmark procedure with bandwidth  $h_l$  ( $l = 1, \dots, 6$ ).

|           | $\mathcal{T}_{\text{MS}}$ | $\mathcal{T}_{\text{bmk}, 1}$ | $\mathcal{T}_{\text{bmk}, 2}$ | $\mathcal{T}_{\text{bmk}, 3}$ | $\mathcal{T}_{\text{bmk}, 4}$ | $\mathcal{T}_{\text{bmk}, 5}$ | $\mathcal{T}_{\text{bmk}, 6}$ |
|-----------|---------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|
| $T = 100$ | 0.065                     | 0.046                         | 0.017                         | 0.011                         | 0.004                         | 0.002                         | 0.003                         |
| $T = 250$ | 0.476                     | 0.439                         | 0.222                         | 0.056                         | 0.018                         | 0.006                         | 0.006                         |
| $T = 500$ | 0.993                     | 0.572                         | 0.603                         | 0.255                         | 0.090                         | 0.035                         | 0.025                         |

the number of clusters  $N$ , we assume  $N = 3$  to be known in what follows.

We implement our clustering algorithm in exactly the same way as in Section A.4 (with  $N = 3$  known). To run the benchmark algorithm in practice, we need to discretize the supremum over all time points  $u \in [0, 1]$  in Step 2 of the algorithm. To make the comparison as fair as possible, we replace this supremum by the maximum over all points  $u \in U_T$ , where  $U_T$  is the grid of locations defined in (A.2) that underlies the location-scale grid  $\mathcal{G}_T = U_T \times H_T$  of our multiscale test. Moreover, we need to choose a bandwidth to run the benchmark algorithm. We consider six different choices:  $h_l = h_{\min} + c_l(h_{\max} - h_{\min})$  with  $c_l = 0.2(l - 1)$  for  $l = 1, \dots, 6$ .

The simulation results are reported in Table A.13. The entries of the table give the empirical probabilities with which the estimated group structure equals the true one  $\{G_1, G_2, G_3\}$ . The two main results are as follows: (i) The performance of the benchmark procedure is heavily influenced by the choice of bandwidth. (ii) Our multiscale method outperforms the benchmark with all considered bandwidth choices  $h_l$ . Results (i) and (ii) nicely illustrate the main advantage of our multiscale method: it automatically selects the “right” scale(s) or bandwidth(s), i.e., those which are most suitable for comparing the trend curves.

## B Application to GDP growth data

In what follows, we revisit an application example from Zhang et al. (2012). The aim is to test the hypothesis of a common trend in the GDP growth time series of several OECD

countries. Since we do not have access to the original dataset of Zhang et al. (2012) and we do not know the exact data specifications used there, we work with data from the following common sources: Refinitiv Datastream, the OECD.Stat database, Federal Reserve Economics Data (FRED) and the Barro-Lee Educational Attainment dataset (Barro and Lee, 2013). We consider a data specification that is as close as possible to the one in Zhang et al. (2012) with one important distinction: In the original study, the authors examined 16 OECD countries (not specifying which ones) over the time period from the fourth quarter of 1975 up to and including the third quarter of 2010, whereas we consider only 11 countries (Australia, Austria, Canada, Finland, France, Germany, Japan, Norway, Switzerland, UK and USA) over the same time span. The reason is that we have access to data of good quality only for these 11 countries. In the following list, we specify the data for our analysis.<sup>6</sup>

- **Gross domestic product (*GDP*):** We use freely available data on *Gross Domestic Product – Expenditure Approach* from the OECD.Stat database (<https://stats.oecd.org/Index.aspx>). To be as close as possible to the specification of the data in Zhang et al. (2012), we use seasonally adjusted quarterly data on GDP expressed in millions of 2015 US dollars.<sup>7</sup>
- **Capital (*K*):** We use data on *Gross Fixed Capital Formation* from the OECD.Stat database. The data are at a quarterly frequency, seasonally adjusted, and expressed in millions of 2015 US dollars. In contrast to Zhang et al. (2012), who use data on *Capital Stock at Constant National Prices*, we choose to work with gross fixed capital formation due to data availability. It is worth noting that since accurate data on capital stock is notoriously difficult to collect, the use of gross fixed capital formation as a measure of capital is standard in the literature; see e.g. Sharma and Dhakal (1994), Lee and Huang

<sup>6</sup>All data were accessed and downloaded on 7 December 2021.

<sup>7</sup>Since the publication of Zhang et al. (2012), the OECD reference year has changed from 2005 to 2015. We have decided to analyse the latest version of the data in order to be able to make more accurate and up-to-date conclusions.



(2002) and Lee (2005).

- **Labour ( $L$ ):** We collect data on the *Number of Employed People* from various sources. For most of the countries (Austria, Australia, Canada, Germany, Japan, UK and USA), we download the OECD data on *Employed Population: Aged 15 and Over* retrieved from FRED (<https://fred.stlouisfed.org/>). The data for France and Switzerland were downloaded from Refinitiv Datastream. For all of the aforementioned countries, the observations are at a quarterly frequency and seasonally adjusted. The data for Finland and Norway were also obtained via Refinitiv Datastream, however, the only quarterly time series that are long enough for our purposes are not seasonally adjusted. Hence, for these two countries, we perform the seasonal adjustment ourselves. We in particular use the default method of the function `seas` from the R package `seasonal` (Sax and Eddelbuettel, 2018) which is an interface to X-13-ARIMA-SEATS, the seasonal adjustment software used by the US Census Bureau. For all of the countries, the data are given in thousands of persons.
- **Human capital ( $H$ ):** We use *Educational Attainment for Population Aged 25 and Over* collected from <http://www.barrolee.com> as a measure of human capital. Since the only available data is five-year census data, we follow Zhang et al. (2012) and use linear interpolation between the observations and constant extrapolation on the boundaries (second and third quarters of 2010) to obtain the quarterly time series.

For each of the  $n = 11$  countries in our sample, we thus observe a quarterly time series  $\mathcal{T}_i = \{(Y_{it}, \mathbf{X}_{it}) : 1 \leq t \leq T\}$  of length  $T = 140$ , where  $Y_{it} = \Delta \ln GDP_{it} := \ln GDP_{it} - \ln GDP_{i(t-1)}$  and  $\mathbf{X}_{it} = (\Delta \ln L_{it}, \Delta \ln K_{it}, \Delta \ln H_{it})^\top$  with  $\Delta \ln L_{it} := \ln L_{it} - \ln L_{i(t-1)}$ ,  $\Delta \ln K_{it} := \ln K_{it} - \ln K_{i(t-1)}$  and  $\Delta \ln H_{it} := \ln H_{it} - \ln H_{i(t-1)}$ . Without loss of generality, we let  $\Delta \ln GDP_{i1} = \Delta \ln L_{i1} = \Delta \ln K_{i1} = \Delta \ln H_{i1} = 0$ . Each time series  $\mathcal{T}_i$  is assumed to

follow the model  $Y_{it} = m_i(t/T) + \beta_i^\top \mathbf{X}_{it} + \alpha_i + \varepsilon_{it}$ , or equivalently,

$$\Delta \ln GDP_{it} = m_i\left(\frac{t}{T}\right) + \beta_{i,1} \Delta \ln L_{it} + \beta_{i,2} \Delta \ln K_{it} + \beta_{i,3} \Delta \ln H_{it} + \alpha_i + \varepsilon_{it} \quad (\text{B.1})$$

for  $1 \leq t \leq T$ , where  $\beta_i = (\beta_{i,1}, \beta_{i,2}, \beta_{i,3})^\top$  is a vector of unknown parameters,  $m_i$  is a country-specific nonparametric time trend and  $\alpha_i$  is a country-specific fixed effect.

In order to test the null hypothesis  $H_0 : m_1 = \dots = m_n$  with  $n = 11$  in model (B.1), we implement our multiscale test as follows:

- We choose  $K$  to be the Epanechnikov kernel and consider the set of location-scale points  $\mathcal{G}_T = U_T \times H_T$ , where  $U_T = \{u \in [0, 1] : u = \frac{8t+1}{2T} \text{ for some } t \in \mathbb{N}\}$  and  $H_T = \{h \in [\frac{\log T}{T}, \frac{1}{4}] : h = \frac{4t}{T} \text{ for some } t \in \mathbb{N}\}$ . We thus take into account all locations  $u$  on an equidistant grid  $U_T$  with step length  $4/T$  and all scales  $h = 4/T, 8/T, 12/T, \dots$  with  $\log T/T \leq h \leq 1/4$ . The choice of the grid  $\mathcal{G}_T$  is motivated by the quarterly frequency of the data: each interval  $\mathcal{I}_{u,h}$  with  $(u, h) \in \mathcal{G}_T$  spans 8, 16, 24,  $\dots$  quarters, i.e., 2, 4, 6,  $\dots$  years. The lower bound  $\log T/T$  on the scales  $h$  in  $H_T$  is motivated by Assumption (C11).
- To obtain an estimator  $\hat{\sigma}_i^2$  of the long-run error variance  $\sigma_i^2$  for each  $i$ , we assume that the error process  $\mathcal{E}_i$  follows an  $\text{AR}(p_i)$  model and apply the difference-based procedure of Khismatullina and Vogt (2020) to the augmented time series  $\{\hat{Y}_{it} : 1 \leq t \leq T\}$  with  $\hat{Y}_{it} = Y_{it} - \hat{\beta}_i^\top \mathbf{X}_{it} - \hat{\alpha}_i$ . We set the tuning parameters  $q$  and  $r$  of the procedure to 20 and 10, respectively, and choose the AR order  $p_i$  by minimizing BIC, which yields  $p_i = 3$  for Australia, Canada and the UK and  $p_i = 1$  for all other countries.<sup>8</sup>
- The critical values  $q_{n,T}(\alpha)$  are computed by Monte Carlo methods as described in Section 3.4, where we set  $L = 5000$ .

Besides these choices, we construct and implement the multiscale test exactly as described in Section 3.

The thus implemented multiscale test rejects the global null hypothesis  $H_0$  at the usual

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<sup>8</sup>We also calculated the values of other information criteria such as FPE, AIC and HQ which, in most of the cases, resulted in the same values of  $p_i$ .

significance levels  $\alpha = 0.01, 0.05, 0.1$ . This result is in line with the findings in Zhang et al. (2012) where the null hypothesis of a common trend is rejected at level  $\alpha = 0.1$ . The main advantage of our multiscale test over the method in Zhang et al. (2012) is that it is much more informative. In particular, it provides information about *which* of the  $n = 11$  countries have different trends and *where* the trends differ. This information is presented graphically in Figures B.1–B.11. Each figure corresponds to a specific pair of countries  $(i, j)$  and is divided into three panels (a)–(c). Panel (a) shows the augmented time series  $\{\hat{Y}_{it} : 1 \leq t \leq T\}$  and  $\{\hat{Y}_{jt} : 1 \leq t \leq T\}$  for the two countries  $i$  and  $j$  that are compared. Panel (b) presents smoothed versions of the time series from (a), in particular, it shows local linear kernel estimates of the two trend functions  $m_i$  and  $m_j$ , where the bandwidth is set to 14 quarters (that is, to 0.1 in terms of rescaled time) and an Epanechnikov kernel is used. Panel (c) presents the results produced by our test for the significance level  $\alpha = 0.05$ : it depicts in grey the set  $\mathcal{S}^{[i,j]}(\alpha)$  of all the intervals for which the test rejects the local null  $H_0^{[i,j]}(u, h)$ . The set of minimal intervals  $\mathcal{S}_{\min}^{[i,j]}(\alpha) \subseteq \mathcal{S}^{[i,j]}(\alpha)$  is highlighted in black. According to (4.3), we can make the following simultaneous confidence statement about the intervals plotted in panels (c) of Figures B.1–B.11: we can claim, with confidence of about 95%, that there is a difference between the functions  $m_i$  and  $m_j$  on each of these intervals.

Out of 55 pairwise comparisons, our test detects differences for 11 pairs of countries  $(i, j)$ . These 11 cases are presented in Figures B.1–B.11. In 9 cases (Figures B.1–B.9), one of the involved countries is Norway. Inspecting the trend estimates in panels (b) of Figures B.1–B.9, the Norwegian trend estimate can be seen to exhibit a strong downward movement at the end of the observation period, whereas the other trend estimates show a much less pronounced downward movement (or even a slight upward movement). According to our test, this is a significant difference between the Norwegian and the other trend functions rather than an artefact of the sampling noise: In all 9 cases, the test rejects the local null

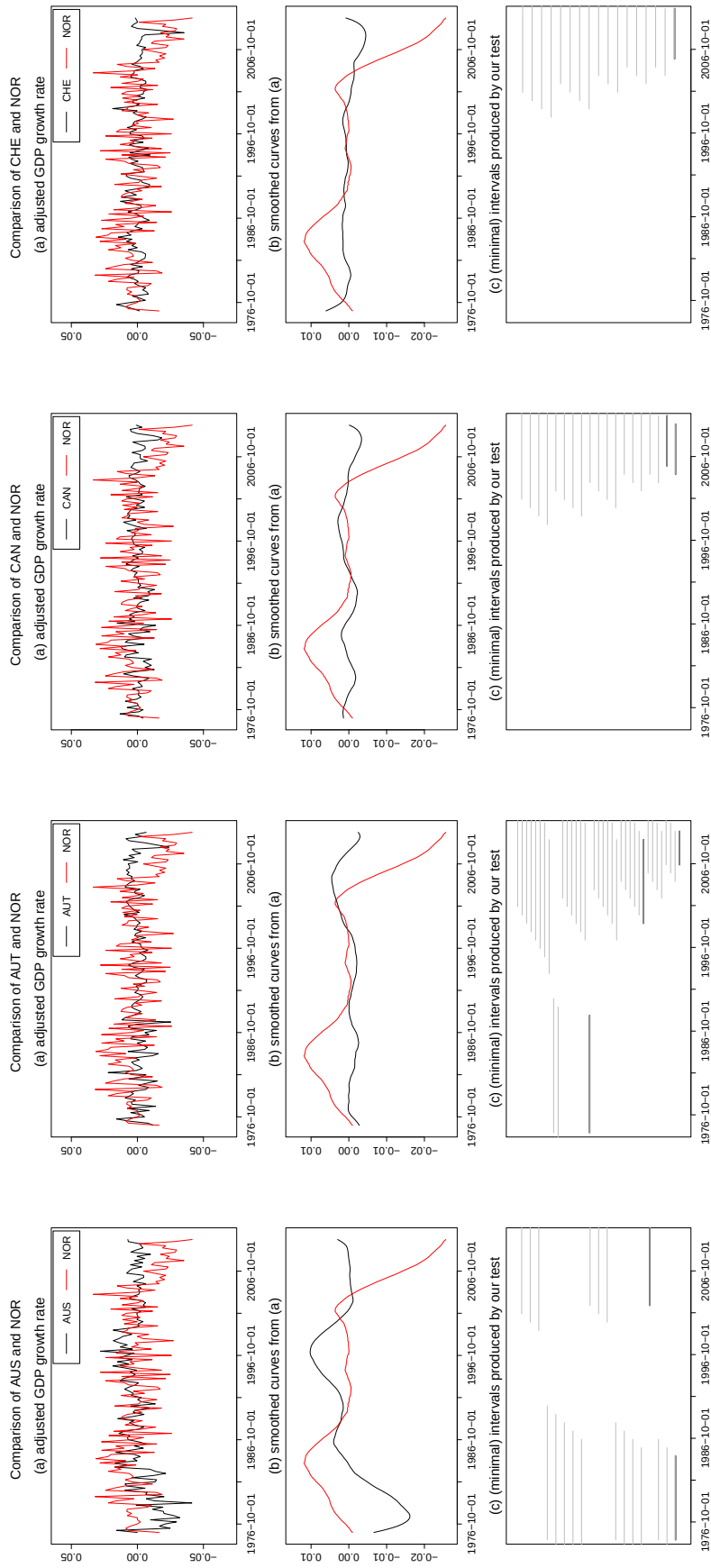


Figure B.1: Test results for the comparison of Australia and Norway. Figure B.2: Test results for the comparison of Austria and Norway. Figure B.3: Test results for the comparison of Canada and Norway. Figure B.4: Test results for the comparison of Switzerland and Norway.

Note: In each figure, panel (a) shows the two augmented time series, panel (b) presents smoothed versions of the augmented time series, and panel (c) depicts the set of intervals  $\mathcal{S}^{[i,j]}(\alpha)$  in grey and the subset of minimal intervals  $\mathcal{S}_{\min}^{[i,j]}(\alpha)$  in black.

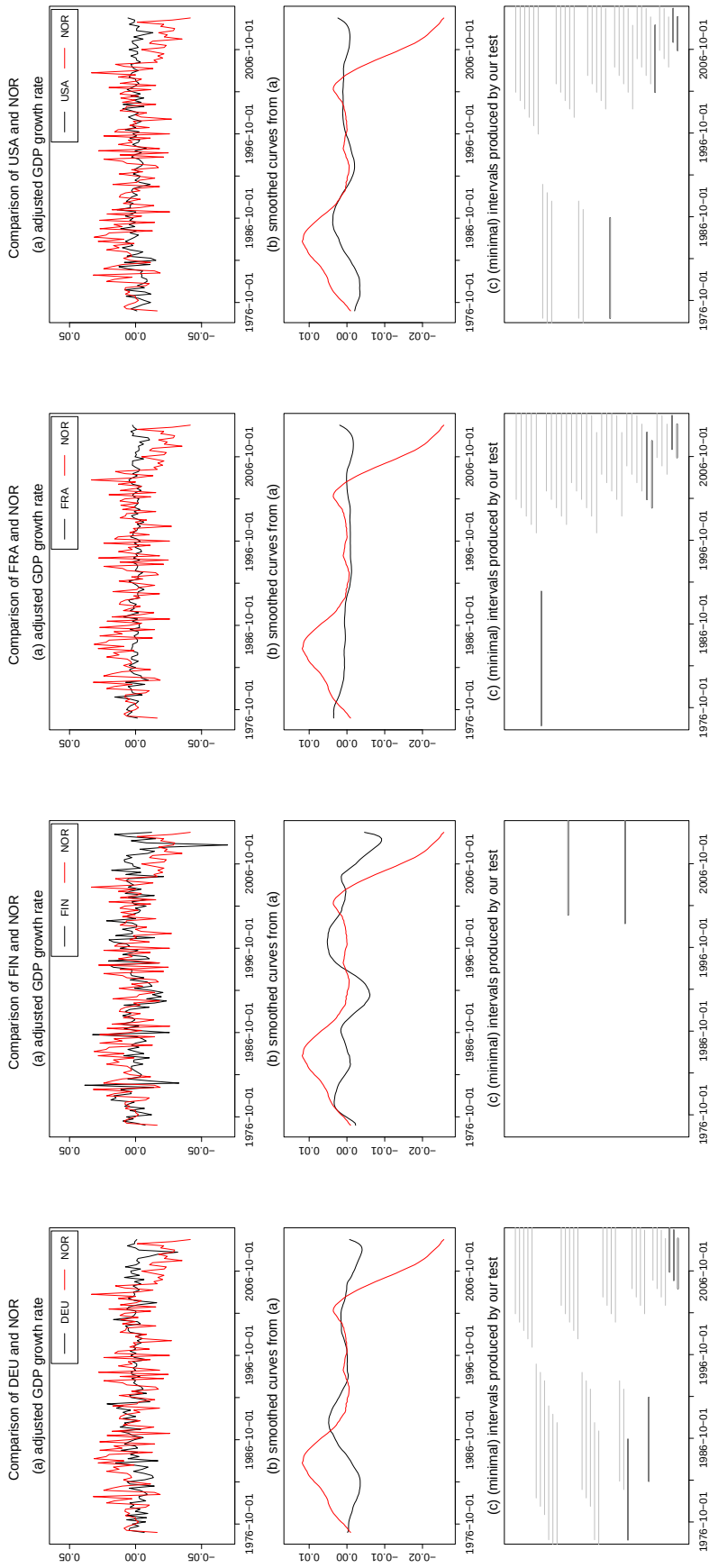


Figure B.5: Test results for the comparison of Germany and Norway. way. Figure B.6: Test results for the comparison of Finland and Norway. way. Figure B.7: Test results for the comparison of France and Norway. way. Figure B.8: Test results for the comparison of the USA and Norway.

Note: In each figure, panel (a) shows the two augmented time series, panel (b) presents smoothed versions of the augmented time series, and panel (c) depicts the set of intervals  $\mathcal{S}^{[i,j]}(\alpha)$  in grey and the subset of minimal intervals  $\mathcal{S}_{\min}^{[i,j]}(\alpha)$  in black.

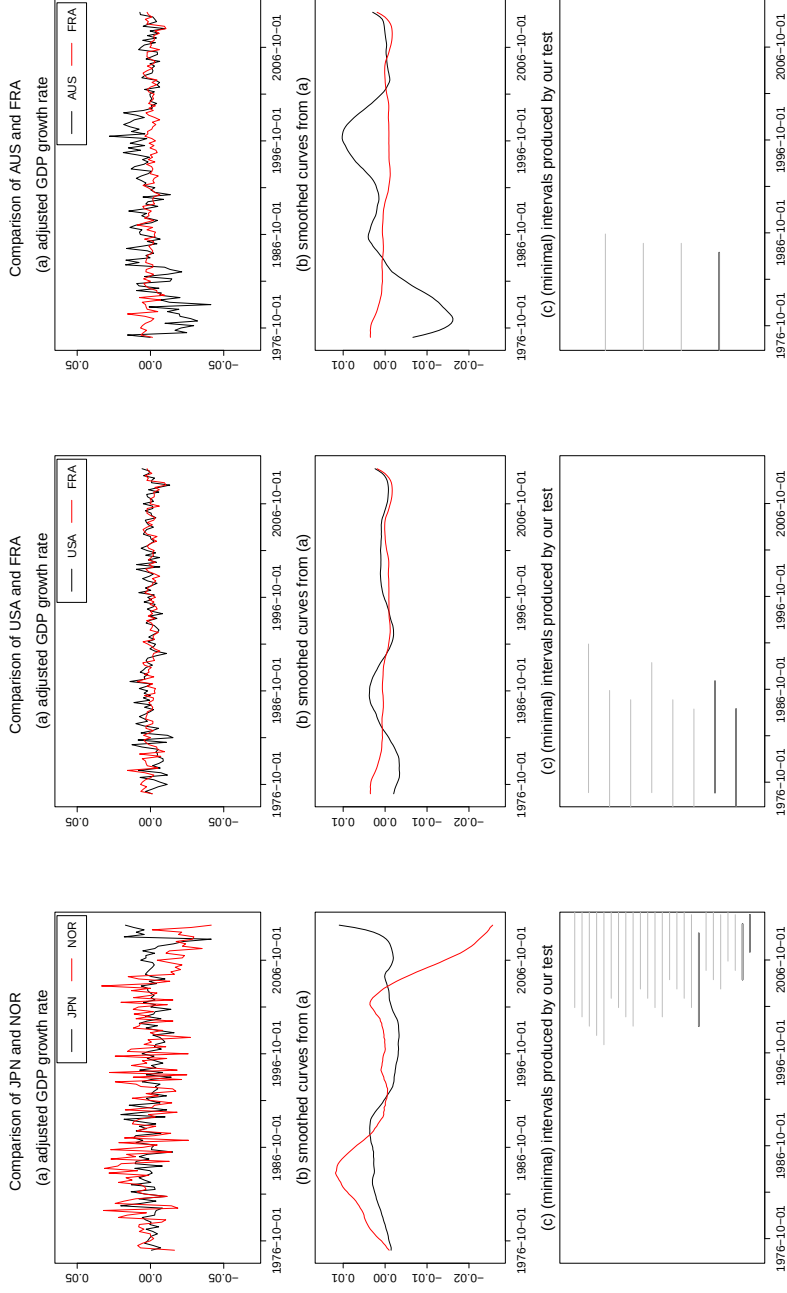


Figure B.9: Test results for the comparison of Japan and Norway. Figure B.10: Test results for the comparison of the USA and France. Figure B.11: Test results for the comparison of Australia and France.

Note: In each figure, panel (a) shows the two augmented time series, panel (b) presents smoothed versions of the augmented time series, and panel (c) depicts the set of intervals  $\mathcal{S}^{[i,j]}(\alpha)$  in grey and the subset of minimal intervals  $\mathcal{S}_{\min}^{[i,j]}(\alpha)$  in black.

for at least one interval which covers the last 10 years of the analysed time period (from the first quarter in 2000 up to the third quarter in 2010). Apart from these differences at the end of the sampling period, our test also finds differences in the beginning, however, only for part of the pairwise comparisons.

Figures B.10 and B.11 present the results of the pairwise comparison between Australia and France and between the USA and France, respectively. In both cases, our test detects differences between the GDP trends only in the beginning of the considered time period. In the case of Australia and France, it is clearly visible in the raw data (panel (a) in Figure B.11) that there is a difference between the trends, whereas this is not so obvious in the case of the USA and France. According to our test, there are indeed significant differences in both cases. In particular, we can claim with confidence at least 95%, that there are differences between the trends of the USA and France (of Australia and France) up to the fourth quarter in 1991 (the fourth quarter in 1986), but there is no evidence of any differences between the trends from 1992 (1987) onwards.

We next apply our clustering techniques to find groups of countries that have the same time trend. We implement our HAC algorithm with  $\alpha = 0.05$  and the same choices as detailed above. The dendrogram that depicts the clustering results is plotted in Figure B.12. The number of clusters is estimated to be  $\hat{N} = 3$ . The rectangles in Figure B.12 indicate the  $\hat{N} = 3$  clusters. In particular, each rectangle is drawn around the branches of the dendrogram that correspond to one of the three clusters. Figure B.13 depicts local linear kernel estimates of the  $n = 11$  GDP time trends (calculated from the augmented time series  $\hat{Y}_{it}$  with bandwidth 0.1 and Epanechnikov kernel). Their colour indicates which cluster they belong to.

The results in Figures B.12 and B.13 show that there is one cluster which consists only of Norway (plotted in red). As we have already discussed above and as becomes apparent from Figure B.13, the Norwegian trend exhibits a strong downward movement at the end

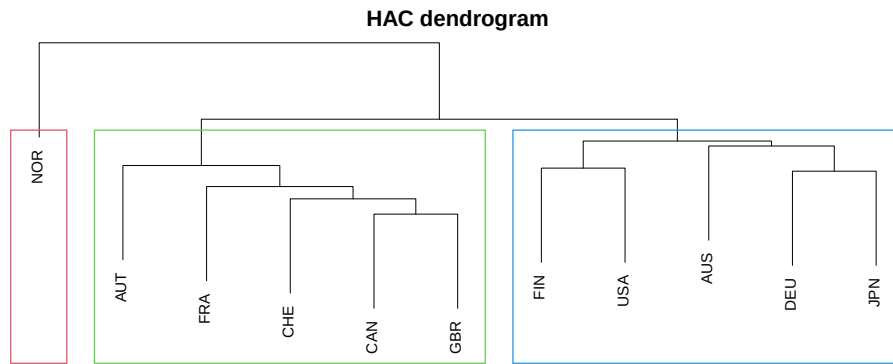


Figure B.12: Dendrogram of the HAC algorithm. Each coloured rectangle corresponds to one of the clusters.

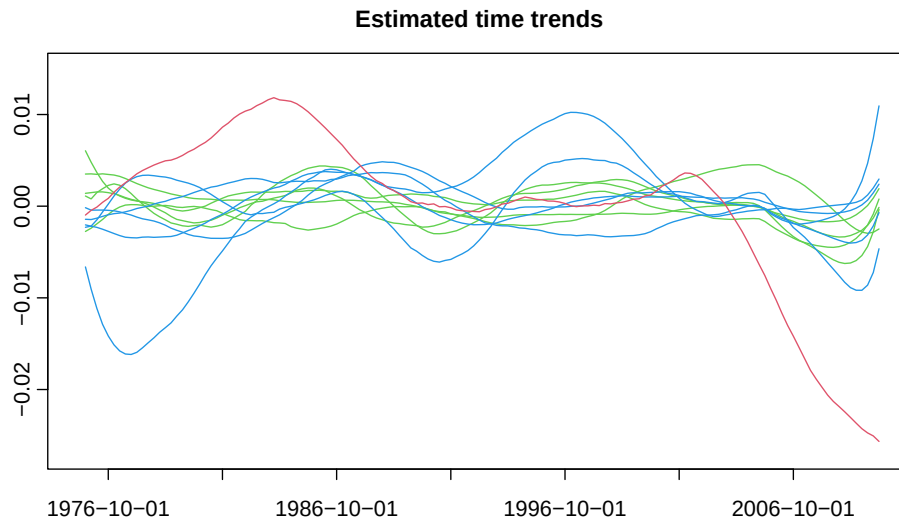


Figure B.13: Local linear estimates of the  $n = 11$  time trends (calculated from the augmented time series  $\hat{Y}_{it}$  with bandwidth  $h = 0.1$  and Epanechnikov kernel). Each trend estimate is coloured according to the cluster that it is assigned to.

of the sampling period, whereas the other trends show a much more moderate downward movement (if at all). This is presumably the reason why the clustering procedure puts Norway in a separate cluster. The algorithm further finds two other clusters, one consisting of the 5 countries Australia, Finland, Germany, Japan and the USA (plotted in blue in Figures B.12 and B.13) and the other one consisting of the 5 countries Austria, Canada, France, Switzerland and the UK (plotted in green in Figures B.12 and B.13). Visual inspection of the trend estimates in Figure B.13 suggests that the GDP time trends in the blue cluster exhibit more pronounced decreases and increases than the GDP time trends in



the green cluster. Hence, overall, the clustering procedure appears to produce a reasonable grouping of the GDP trends.

## C Technical details

In what follows, we prove the theoretical results from Sections 4 and 5. We use the following notation: The symbol  $C$  denotes a universal real constant which may take a different value on each occurrence. For  $a, b \in \mathbb{R}$ , we write  $a \vee b = \max\{a, b\}$ . For  $x \in \mathbb{R}_{\geq 0}$ , we let  $\lfloor x \rfloor$  denote the integer value of  $x$  and  $\lceil x \rceil$  the smallest integer greater than or equal to  $x$ . For any set  $A$ , the symbol  $|A|$  denotes the cardinality of  $A$ . The expression  $X \stackrel{\mathcal{D}}{=} Y$  means that the two random variables  $X$  and  $Y$  have the same distribution. Finally, we sometimes use the notation  $a_T \ll b_T$  to express that  $a_T = o(b_T)$ .

### Auxiliary results

Let  $\{Z_t\}_{t=-\infty}^{\infty}$  be a stationary time series process with  $Z_t \in \mathcal{L}^q$  for some  $q > 2$  and  $\mathbb{E}[Z_t] = 0$ . Assume that  $Z_t$  can be represented as  $Z_t = g(\dots, \eta_{t-1}, \eta_t)$ , where  $\eta_t$  are i.i.d. variables and  $g : \mathbb{R}^{\infty} \rightarrow \mathbb{R}$  is a measurable function. We first state a Nagaev-type inequality from Wu and Wu (2016).

**Definition C.1.** Let  $q > 0$  and  $\alpha > 0$ . The dependence adjusted norm of the process  $Z. = \{Z_t\}_{t=-\infty}^{\infty}$  is given by  $\|Z.\|_{q,\alpha} = \sup_{t \geq 0} (t+1)^{\alpha} \sum_{s=t}^{\infty} \delta_q(g, s)$ .

**Proposition C.2** (Wu and Wu (2016), Theorem 2). Assume that  $\|Z.\|_{q,\alpha} < \infty$  with  $q > 2$  and  $\alpha > 1/2 - 1/q$ . Let  $S_T = a_1 Z_1 + \dots + a_T Z_T$ , where  $a_1, \dots, a_T$  are real numbers with  $\sum_{t=1}^T a_t^2 = T$ . Then for any  $w > 0$ ,

$$\mathbb{P}(|S_T| \geq w) \leq C_1 \frac{|a|_q^q \|Z.\|_{q,\alpha}^q}{w^q} + C_2 \exp\left(-\frac{C_3 w^2}{T \|Z.\|_{2,\alpha}^2}\right),$$

where  $C_1, C_2, C_3$  are constants that only depend on  $q$  and  $\alpha$ .

The following lemma is a simple consequence of the above inequality.

**Lemma C.3.** Let  $\sum_{s=t}^{\infty} \delta_q(g, s) = O(t^{-\alpha})$  for some  $q > 2$  and  $\alpha > 1/2 - 1/q$ . Then

$$\frac{1}{\sqrt{T}} \sum_{t=1}^T Z_t = O_p(1).$$

**Proof of Lemma C.3.** Let  $\eta > 0$  be a fixed number. We apply Proposition C.2 to the sum  $S_T = \sum_{t=1}^T a_t Z_t$  with  $a_t = 1$  for all  $t$ . The assumption  $\sum_{s=t}^{\infty} \delta_q(g, s) = O(t^{-\alpha})$  implies that  $\|Z_{\cdot}\|_{2,\alpha} \leq \|Z_{\cdot}\|_{q,\alpha} \leq C_Z < \infty$ . Hence, for  $w$  chosen sufficiently large, we get

$$\begin{aligned} \mathbb{P}\left(\left|\sum_{t=1}^T Z_t\right| \geq \sqrt{T}w\right) &\leq C_1 \frac{TC_Z^q}{T^{q/2}w^q} + C_2 \exp\left(-\frac{C_3Tw^2}{TC_Z^2}\right) \\ &= \frac{\{C_1C_Z^q\}T^{1-q/2}}{w^q} + C_2 \exp\left(-\frac{C_3w^2}{C_Z^2}\right) \leq \eta \end{aligned}$$

for all  $T$ . This means that  $\sum_{t=1}^T Z_t/\sqrt{T} = O_p(1)$ .  $\square$

Let  $\Delta\varepsilon_{it} = \varepsilon_{it} - \varepsilon_{it-1}$  and  $\Delta\mathbf{X}_{it} = \mathbf{X}_{it} - \mathbf{X}_{it-1}$ . By Assumptions (C1) and (C4),  $\Delta\varepsilon_{it} = \Delta g_i(\mathcal{F}_{it})$  and  $\Delta\mathbf{X}_{it} = \Delta\mathbf{h}_i(\mathcal{G}_{it})$ . We further define

$$\begin{aligned} \mathbf{a}_i(\mathcal{H}_{it}) &:= \Delta\mathbf{h}_i(\mathcal{G}_{it})\Delta g_i(\mathcal{F}_{it}) = \Delta\mathbf{X}_{it}\Delta\varepsilon_{it} \\ \mathbf{b}_i(\mathcal{G}_{it}) &:= \Delta\mathbf{h}_i(\mathcal{G}_{it})\Delta\mathbf{h}_i(\mathcal{G}_{it})^\top = \Delta\mathbf{X}_{it}\Delta\mathbf{X}_{it}^\top, \end{aligned}$$

where  $\mathbf{a}_i = (a_{ij})_{j=1}^d$ ,  $\mathbf{b}_i = (b_{ikl})_{k,l=1}^d$  and  $\mathcal{H}_{it} = (\mathcal{H}_{it,1}, \dots, \mathcal{H}_{it,d})^\top$  with  $\mathcal{H}_{it,j} = (\dots, \nu_{it-1,j}, \nu_{it,j})$  and  $\nu_{it,j} = (\eta_{it}, \xi_{it,j})$ . The next result gives bounds on the physical dependence measures of the processes  $\{\mathbf{a}_i(\mathcal{H}_{it})\}_{t=-\infty}^{\infty}$  and  $\{\mathbf{b}_i(\mathcal{G}_{it})\}_{t=-\infty}^{\infty}$ .

**Lemma C.4.** Let Assumptions (C1), (C3), (C4) and (C6) be satisfied. Then for each  $i$ ,  $j$ ,  $k$  and  $l$ , it holds that

$$\begin{aligned} \sum_{s=t}^{\infty} \delta_p(a_{ij}, s) &= O(t^{-\alpha}) \quad \text{for } p = \min\{q, q'\}/2 \text{ and some } \alpha > 1/2 - 1/p \\ \sum_{s=t}^{\infty} \delta_p(b_{ikl}, s) &= O(t^{-\alpha}) \quad \text{for } p = q'/2 \text{ and some } \alpha > 1/2 - 1/p. \end{aligned}$$

**Proof of Lemma C.4.** We only prove the first statement. The second one follows by analogous arguments. By the definition of the physical dependence measure and the Cauchy-

Schwarz inequality, we have with  $p = \min\{q, q'\}/2$  that

$$\begin{aligned}
\delta_p(a_{ij}, s) &= \|a_{ij}(\mathcal{H}_{it,j}) - a_{ij}(\mathcal{H}'_{it,j})\|_p \\
&= \|\Delta h_{ij}(\mathcal{G}_{it})\Delta g_i(\mathcal{F}_{it}) - \Delta h_{ij}(\mathcal{G}'_{it})\Delta g_i(\mathcal{F}'_{it})\|_p \\
&\leq \|h_{ij}(\mathcal{G}_{it})g_i(\mathcal{F}_{it}) - h_{ij}(\mathcal{G}'_{it})g_i(\mathcal{F}'_{it})\|_p \\
&\quad + \|h_{ij}(\mathcal{G}_{it-1})g_i(\mathcal{F}_{it-1}) - h_{ij}(\mathcal{G}'_{it-1})g_i(\mathcal{F}'_{it-1})\|_p \\
&\quad + \|h_{ij}(\mathcal{G}_{it-1})g_i(\mathcal{F}_{it}) - h_{ij}(\mathcal{G}'_{it-1})g_i(\mathcal{F}'_{it})\|_p \\
&\quad + \|h_{ij}(\mathcal{G}_{it})g_i(\mathcal{F}_{it-1}) - h_{ij}(\mathcal{G}'_{it})g_i(\mathcal{F}'_{it-1})\|_p \\
&= \|\{h_{ij}(\mathcal{G}_{it}) - h_{ij}(\mathcal{G}'_{it})\}g_i(\mathcal{F}_{it}) + h_{ij}(\mathcal{G}'_{it})\{g_i(\mathcal{F}_{it}) - g_i(\mathcal{F}'_{it})\}\|_p \\
&\quad + \|\{h_{ij}(\mathcal{G}_{it-1}) - h_{ij}(\mathcal{G}'_{it-1})\}g_i(\mathcal{F}_{it-1}) + h_{ij}(\mathcal{G}'_{it-1})\{g_i(\mathcal{F}_{it-1}) - g_i(\mathcal{F}'_{it-1})\}\|_p \\
&\quad + \|\{h_{ij}(\mathcal{G}_{it-1}) - h_{ij}(\mathcal{G}'_{it-1})\}g_i(\mathcal{F}_{it}) + h_{ij}(\mathcal{G}'_{it-1})\{g_i(\mathcal{F}_{it}) - g_i(\mathcal{F}'_{it})\}\|_p \\
&\quad + \|\{h_{ij}(\mathcal{G}_{it}) - h_{ij}(\mathcal{G}'_{it})\}g_i(\mathcal{F}_{it-1}) + h_{ij}(\mathcal{G}'_{it})\{g_i(\mathcal{F}_{it-1}) - g_i(\mathcal{F}'_{it-1})\}\|_p \\
&\leq \delta_{2p}(h_{ij}, t)\|g_i(\mathcal{F}_t)\|_{2p} + \delta_{2p}(g_i, t)\|h_{ij}(\mathcal{G}'_{it})\|_{2p} \\
&\quad + \delta_{2p}(h_{ij}, t-1)\|g_i(\mathcal{F}_{t-1})\|_{2p} + \delta_{2p}(g_i, t-1)\|h_{ij}(\mathcal{G}'_{it-1})\|_{2p} \\
&\quad + \delta_{2p}(h_{ij}, t-1)\|g_i(\mathcal{F}_t)\|_{2p} + \delta_{2p}(g_i, t)\|h_{ij}(\mathcal{G}'_{it-1})\|_{2p} \\
&\quad + \delta_{2p}(h_{ij}, t)\|g_i(\mathcal{F}_{t-1})\|_{2p} + \delta_{2p}(g_i, t-1)\|h_{ij}(\mathcal{G}'_{it})\|_{2p},
\end{aligned}$$

where  $\mathcal{H}'_{it,j} = (\dots, \nu_{i(-1),j}, \nu'_{i0,j}, \nu_{i1,j}, \dots, \nu_{it-1,j}, \nu_{it,j})$ ,  $\mathcal{G}'_{it,j} = (\dots, \xi_{i(-1),j}, \xi'_{i0,j}, \xi_{i1,j}, \dots, \xi_{it-1,j}, \xi_{it,j})$  and  $\mathcal{F}'_{it} = (\dots, \eta_{i(-1)}, \eta'_{i0}, \eta_{i1}, \dots, \eta_{it-1}, \eta_{it})$  are coupled processes with  $\nu'_{i0,j}$ ,  $\xi'_{i0,j}$  and  $\eta'_{i0,j}$  being i.i.d. copies of  $\nu_{i0,j}$ ,  $\xi_{i0,j}$  and  $\eta_{i0}$ . From this and Assumptions (C1), (C3), (C4) and (C6), it immediately follows that  $\sum_{s=t}^{\infty} \delta_p(a_{ij}, s) = O(t^{-\alpha})$ .  $\square$

We now show that the estimator  $\hat{\beta}_i$  is  $\sqrt{T}$ -consistent for each  $i$  under our conditions.

**Lemma C.5.** *Let Assumptions (C1), (C3) and (C4)–(C7) be satisfied. Then for each  $i$ ,*

$$\hat{\beta}_i - \beta_i = O_p\left(\frac{1}{\sqrt{T}}\right).$$

**Proof of Lemma C.5.** The estimator  $\widehat{\beta}_i$  can be written as

$$\begin{aligned}\widehat{\beta}_i &= \left( \sum_{t=2}^T \Delta \mathbf{X}_{it} \Delta \mathbf{X}_{it}^\top \right)^{-1} \sum_{t=2}^T \Delta \mathbf{X}_{it} \Delta Y_{it} \\ &= \left( \sum_{t=2}^T \Delta \mathbf{X}_{it} \Delta \mathbf{X}_{it}^\top \right)^{-1} \sum_{t=2}^T \Delta \mathbf{X}_{it} \left( \Delta \mathbf{X}_{it}^\top \beta_i + \Delta m_{it} + \Delta \varepsilon_{it} \right) \\ &= \beta_i + \left( \sum_{t=2}^T \Delta \mathbf{X}_{it} \Delta \mathbf{X}_{it}^\top \right)^{-1} \sum_{t=2}^T \Delta \mathbf{X}_{it} \Delta m_{it} + \left( \sum_{t=2}^T \Delta \mathbf{X}_{it} \Delta \mathbf{X}_{it}^\top \right)^{-1} \sum_{t=2}^T \Delta \mathbf{X}_{it} \Delta \varepsilon_{it},\end{aligned}$$

where  $\Delta \mathbf{X}_{it} = \mathbf{X}_{it} - \mathbf{X}_{it-1}$ ,  $\Delta \varepsilon_{it} = \varepsilon_{it} - \varepsilon_{it-1}$  and  $\Delta m_{it} = m_i(\frac{t}{T}) - m_i(\frac{t-1}{T})$ . Hence,

$$\begin{aligned}\sqrt{T}(\widehat{\beta}_i - \beta_i) &= \left( \frac{1}{T} \sum_{t=2}^T \Delta \mathbf{X}_{it} \Delta \mathbf{X}_{it}^\top \right)^{-1} \frac{1}{\sqrt{T}} \sum_{t=2}^T \Delta \mathbf{X}_{it} \Delta m_{it} \\ &\quad + \left( \frac{1}{T} \sum_{t=2}^T \Delta \mathbf{X}_{it} \Delta \mathbf{X}_{it}^\top \right)^{-1} \frac{1}{\sqrt{T}} \sum_{t=2}^T \Delta \mathbf{X}_{it} \Delta \varepsilon_{it}.\end{aligned}\tag{C.1}$$

In what follows, we show that

$$\frac{1}{\sqrt{T}} \sum_{t=2}^T \Delta \mathbf{X}_{it} \Delta \varepsilon_{it} = O_p(1)\tag{C.2}$$

$$\left( \frac{1}{T} \sum_{t=2}^T \Delta \mathbf{X}_{it} \Delta \mathbf{X}_{it}^\top \right)^{-1} = O_p(1)\tag{C.3}$$

$$\frac{1}{\sqrt{T}} \sum_{t=2}^T \Delta \mathbf{X}_{it} \Delta m_{it} = O_p\left(\frac{1}{\sqrt{T}}\right).\tag{C.4}$$

Lemma C.5 follows from applying these three statements together with standard arguments to formula (C.1).

Since  $\mathbb{E}[\Delta \mathbf{X}_{it} \Delta \varepsilon_{it}] = 0$  by (C7) and  $\sum_{s=t}^\infty \delta_p(a_{ij}, s) = O(t^{-\alpha})$  for some  $p > 2$ ,  $\alpha > 1/2 - 1/p$  and all  $j$  by Lemma C.4, the claim (C.2) follows upon applying Lemma C.3. Another application of Lemma C.3 yields that

$$\frac{1}{T} \sum_{t=2}^T \left\{ \Delta \mathbf{X}_{it} \Delta \mathbf{X}_{it}^\top - \mathbb{E}[\Delta \mathbf{X}_{it} \Delta \mathbf{X}_{it}^\top] \right\} = O_p\left(\frac{1}{\sqrt{T}}\right).$$

As  $\mathbb{E}[\Delta \mathbf{X}_{it} \Delta \mathbf{X}_{it}^\top]$  is invertible, we can invoke Slutsky's lemma to obtain (C.3). By assumption,  $m_i$  is Lipschitz continuous, which implies that  $|\Delta m_{it}| = |m_i(\frac{t}{T}) - m_i(\frac{t-1}{T})| \leq C/T$  for all  $t \in \{1, \dots, T\}$  and some constant  $C > 0$ . Hence,

$$\left| \frac{1}{\sqrt{T}} \sum_{t=2}^T \Delta X_{it,j} \Delta m_{it} \right| \leq \frac{1}{\sqrt{T}} \sum_{t=2}^T |\Delta X_{it,j}| \cdot |\Delta m_{it}|$$

$$\leq \frac{C}{\sqrt{T}} \cdot \frac{1}{T} \sum_{t=2}^T |\Delta X_{it,j}| = O_p\left(\frac{1}{\sqrt{T}}\right),$$

where we have used that  $T^{-1} \sum_{t=2}^T |\Delta X_{it,j}| = O_p(1)$  by Markov's inequality. This yields (C.4).  $\square$

**Lemma C.6.** *Let*

$$\hat{\sigma}_i^2 = \frac{1}{2(M-1)s_T} \sum_{m=1}^M \left[ \sum_{t=1}^{s_T} \left( Y_{i(t+ms_T)} - Y_{i(t+(m-1)s_T)} - \hat{\beta}_i^\top (\mathbf{X}_{i(t+ms_T)} - \mathbf{X}_{i(t+(m-1)s_T)}) \right) \right]^2,$$

be the subseries estimator of  $\sigma_i^2$ , where  $s_T \asymp T^{1/3}$  is the length of the subseries and  $M = \lfloor T/s_T \rfloor$  is the largest integer not exceeding  $T/s_T$ . Under Assumptions (C1)–(C7),

$$\hat{\sigma}_i^2 = \sigma_i^2 + O_p(T^{-1/3})$$

for each  $i$ .

**Proof of Lemma C.6.** Let  $Y_{it}^\circ := m_i(t/T) + \varepsilon_{it}$ . Using simple arithmetic calculations, we can rewrite  $\hat{\sigma}_i^2$  as  $\hat{\sigma}_i^2 = \hat{\sigma}_{i,A}^2 + \hat{\sigma}_{i,B}^2 - \hat{\sigma}_{i,C}^2$ , where

$$\begin{aligned} \hat{\sigma}_{i,A}^2 &= \frac{1}{2(M-1)s_T} \sum_{m=1}^M \left[ \sum_{t=1}^{s_T} (Y_{i(t+ms_T)}^\circ - Y_{i(t+(m-1)s_T)}^\circ) \right]^2 \\ \hat{\sigma}_{i,B}^2 &= \frac{1}{2(M-1)s_T} \sum_{m=1}^M \left[ \sum_{t=1}^{s_T} (\hat{\beta}_i - \beta_i)^\top (\mathbf{X}_{i(t+ms_T)} - \mathbf{X}_{i(t+(m-1)s_T)}) \right]^2 \\ \hat{\sigma}_{i,C}^2 &= \frac{1}{(M-1)s_T} \sum_{m=1}^M \left[ \sum_{t=1}^{s_T} (Y_{i(t+ms_T)}^\circ - Y_{i(t+(m-1)s_T)}^\circ) \right. \\ &\quad \left. \times \sum_{t=1}^{s_T} (\hat{\beta}_i - \beta_i)^\top (\mathbf{X}_{i(t+ms_T)} - \mathbf{X}_{i(t+(m-1)s_T)}) \right]. \end{aligned}$$

By Carlstein (1986) and Wu and Zhao (2007),  $\hat{\sigma}_{i,A}^2 = \sigma_i^2 + O_p(T^{-1/3})$ . Moreover, under our assumptions, it is straightforward to see that  $\hat{\sigma}_{i,B}^2 = O_p(T^{-1/3})$  and  $\hat{\sigma}_{i,C}^2 = O_p(T^{-1/3})$ .  $\square$

## Proof of Theorem 4.1

We first summarize the main proof strategy, which splits up into five steps, and then fill in the details. We in particular defer the proofs of some intermediate results to the end of

the section.

### Step 1

To start with, we consider a simplified setting where the parameter vectors  $\beta_i$  are known. In this case, we can replace the estimators  $\hat{\beta}_i$  in the definition of the statistic  $\hat{\Phi}_{n,T}$  by the true vectors  $\beta_i$  themselves. This leads to the simpler statistic

$$\hat{\Phi}_{n,T} = \max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} \left\{ \left| \frac{\hat{\phi}_{ij,T}(u,h)}{\{\hat{\sigma}_i^2 + \hat{\sigma}_j^2\}^{1/2}} \right| - \lambda(h) \right\},$$

where

$$\hat{\phi}_{ij,T}(u,h) = \sum_{t=1}^T w_{t,T}(u,h) \{(\varepsilon_{it} - \bar{\varepsilon}_i) - (\varepsilon_{jt} - \bar{\varepsilon}_j)\}$$

and  $\hat{\sigma}_i^2$  is computed in exactly the same way as  $\hat{\sigma}_i^2$  except that all occurrences of  $\hat{\beta}_i$  are replaced by  $\beta_i$ . By assumption,  $\hat{\sigma}_i^2 = \sigma_i^2 + o_p(\rho_T)$  with  $\rho_T = o(1/\log T)$ . For most estimators of  $\sigma_i^2$  including those discussed in Section 4, this assumption immediately implies that  $\hat{\sigma}_i^2 = \sigma_i^2 + o_p(\rho_T)$  as well. In the sequel, we thus take for granted that the estimator  $\hat{\sigma}_i^2$  has this property.

We now have a closer look at the statistic  $\hat{\Phi}_{n,T}$ . We in particular show that there exists an identically distributed version  $\tilde{\Phi}_{n,T}$  of  $\hat{\Phi}_{n,T}$  which is close to the Gaussian statistic  $\Phi_{n,T}$  from (3.10). More formally, we prove the following result.

**Proposition C.7.** *There exist statistics  $\{\tilde{\Phi}_{n,T} : T = 1, 2, \dots\}$  with the following two properties: (i)  $\tilde{\Phi}_{n,T}$  has the same distribution as  $\hat{\Phi}_{n,T}$  for any  $T$ , and (ii)*

$$|\tilde{\Phi}_{n,T} - \Phi_{n,T}| = o_p(\delta_T),$$

where  $\delta_T = T^{1/q}/\sqrt{Th_{\min}} + \rho_T\sqrt{\log T}$  and  $\Phi_{n,T}$  is a Gaussian statistic as defined in (3.10).

The proof makes heavy use of strong approximation theory for dependent processes. As it is quite technical, it is postponed to the end of this section.

## Step 2

In this step, we establish some properties of the Gaussian statistic  $\Phi_{n,T}$ . Specifically, we prove the following result.

**Proposition C.8.** *It holds that*

$$\sup_{x \in \mathbb{R}} \mathbb{P}(|\Phi_{n,T} - x| \leq \delta_T) = o(1),$$

where  $\delta_T = T^{1/q}/\sqrt{Th_{\min}} + \rho_T\sqrt{\log T}$ .

Roughly speaking, this proposition says that the random variable  $\Phi_{n,T}$  does not concentrate too strongly in small regions of the form  $[x - \delta_T, x + \delta_T]$  with  $\delta_T$  converging to 0. The main technical tool for deriving it are anti-concentration bounds for Gaussian random vectors. The details are provided below.

## Step 3

We now use Steps 1 and 2 to prove that

$$\sup_{x \in \mathbb{R}} |\mathbb{P}(\widehat{\Phi}_{n,T} \leq x) - \mathbb{P}(\Phi_{n,T} \leq x)| = o(1). \quad (\text{C.5})$$

**Proof of (C.5).** It holds that

$$\begin{aligned} & \sup_{x \in \mathbb{R}} \left| \mathbb{P}(\widehat{\Phi}_{n,T} \leq x) - \mathbb{P}(\Phi_{n,T} \leq x) \right| \\ &= \sup_{x \in \mathbb{R}} \left| \mathbb{P}(\widetilde{\Phi}_{n,T} \leq x) - \mathbb{P}(\Phi_{n,T} \leq x) \right| \\ &= \sup_{x \in \mathbb{R}} \left| \mathbb{E} \left[ \mathbb{1}(\widetilde{\Phi}_{n,T} \leq x) - \mathbb{1}(\Phi_{n,T} \leq x) \right] \right| \\ &\leq \sup_{x \in \mathbb{R}} \left| \mathbb{E} \left[ \left\{ \mathbb{1}(\widetilde{\Phi}_{n,T} \leq x) - \mathbb{1}(\Phi_{n,T} \leq x) \right\} \mathbb{1}(|\widetilde{\Phi}_{n,T} - \Phi_{n,T}| \leq \delta_T) \right] \right| \\ &\quad + \mathbb{E} \left[ \mathbb{1}(|\widetilde{\Phi}_{n,T} - \Phi_{n,T}| > \delta_T) \right]. \end{aligned}$$

Moreover, since

$$\mathbb{E} \left[ \mathbb{1}(|\widetilde{\Phi}_{n,T} - \Phi_{n,T}| > \delta_T) \right] = \mathbb{P}(|\widetilde{\Phi}_{n,T} - \Phi_{n,T}| > \delta_T) = o(1)$$

by Step 1 and

$$\begin{aligned}
& \sup_{x \in \mathbb{R}} \left| \mathbb{E} \left[ \left\{ \mathbb{1}(\tilde{\Phi}_{n,T} \leq x) - \mathbb{1}(\Phi_{n,T} \leq x) \right\} \mathbb{1}(|\tilde{\Phi}_{n,T} - \Phi_{n,T}| \leq \delta_T) \right] \right| \\
& \leq \sup_{x \in \mathbb{R}} \mathbb{E} \left[ \mathbb{1}(|\Phi_{n,T} - x| \leq \delta_T, |\tilde{\Phi}_{n,T} - \Phi_{n,T}| \leq \delta_T) \right] \\
& \leq \sup_{x \in \mathbb{R}} \mathbb{P}(|\Phi_{n,T} - x| \leq \delta_T) = o(1)
\end{aligned}$$

by Step 2, we arrive at (C.5). □

#### Step 4

In this step, we show that the auxiliary statistic  $\widehat{\widehat{\Phi}}_{n,T}$  is close to  $\widehat{\Phi}_{n,T}$  in the following sense.

**Proposition C.9.** *It holds that*

$$\widehat{\widehat{\Phi}}_{n,T} - \widehat{\Phi}_{n,T} = o_p(\delta_T)$$

with  $\delta_T = T^{1/q} / \sqrt{Th_{\min}} + \rho_T \sqrt{\log T}$ .

The proof can be found at the end of this section.

#### Step 5

We finally show that

$$\sup_{x \in \mathbb{R}} |\mathbb{P}(\widehat{\widehat{\Phi}}_{n,T} \leq x) - \mathbb{P}(\widehat{\Phi}_{n,T} \leq x)| = o(1). \quad (\text{C.6})$$

**Proof of (C.6).** To start with, we verify that for any  $x \in \mathbb{R}$  and any  $\delta > 0$ ,

$$\begin{aligned}
& \mathbb{P}(\widehat{\widehat{\Phi}}_{n,T} \leq x - \delta) - \mathbb{P}(|\widehat{\widehat{\Phi}}_{n,T} - \widehat{\Phi}_{n,T}| > \delta) \\
& \leq \mathbb{P}(\widehat{\Phi}_{n,T} \leq x) \leq \mathbb{P}(\widehat{\widehat{\Phi}}_{n,T} \leq x + \delta) + \mathbb{P}(|\widehat{\widehat{\Phi}}_{n,T} - \widehat{\Phi}_{n,T}| > \delta).
\end{aligned} \quad (\text{C.7})$$

It holds that

$$\begin{aligned}
\mathbb{P}(\widehat{\Phi}_{n,T} \leq x) &= \mathbb{P}(\widehat{\Phi}_{n,T} \leq x, |\widehat{\widehat{\Phi}}_{n,T} - \widehat{\Phi}_{n,T}| \leq \delta) + \mathbb{P}(\widehat{\Phi}_{n,T} \leq x, |\widehat{\widehat{\Phi}}_{n,T} - \widehat{\Phi}_{n,T}| > \delta) \\
&\leq \mathbb{P}(\widehat{\Phi}_{n,T} \leq x, \widehat{\Phi}_{n,T} - \delta \leq \widehat{\widehat{\Phi}}_{n,T} \leq \widehat{\Phi}_{n,T} + \delta) + \mathbb{P}(|\widehat{\widehat{\Phi}}_{n,T} - \widehat{\Phi}_{n,T}| > \delta)
\end{aligned}$$



$$\leq \mathbb{P}(\widehat{\Phi}_{n,T} \leq x + \delta) + \mathbb{P}(|\widehat{\Phi}_{n,T} - \Phi_{n,T}| > \delta)$$

and analogously

$$\mathbb{P}(\widehat{\Phi}_{n,T} \leq x - \delta) \leq \mathbb{P}(\Phi_{n,T} \leq x) + \mathbb{P}(|\widehat{\Phi}_{n,T} - \Phi_{n,T}| > \delta).$$

Combining these two inequalities, we arrive at (C.7).

Now let  $x \in \mathbb{R}$  be any point such that  $\mathbb{P}(\widehat{\Phi}_{n,T} \leq x) \geq \mathbb{P}(\Phi_{n,T} \leq x)$ . With the help of (C.7), we get that

$$\begin{aligned} \left| \mathbb{P}(\widehat{\Phi}_{n,T} \leq x) - \mathbb{P}(\Phi_{n,T} \leq x) \right| &= \mathbb{P}(\widehat{\Phi}_{n,T} \leq x) - \mathbb{P}(\Phi_{n,T} \leq x) \\ &\leq \mathbb{P}(\widehat{\Phi}_{n,T} \leq x + \delta_T) + \mathbb{P}(|\widehat{\Phi}_{n,T} - \Phi_{n,T}| > \delta_T) \\ &\quad - \mathbb{P}(\Phi_{n,T} \leq x) \\ &= \mathbb{P}(\widehat{\Phi}_{n,T} \leq x + \delta_T) - \mathbb{P}(\Phi_{n,T} \leq x + \delta_T) \\ &\quad + \mathbb{P}(\Phi_{n,T} \leq x + \delta_T) - \mathbb{P}(\Phi_{n,T} \leq x) \\ &\quad + \mathbb{P}(|\widehat{\Phi}_{n,T} - \Phi_{n,T}| > \delta_T). \end{aligned}$$

Analogously, for any point  $x \in \mathbb{R}$  with  $\mathbb{P}(\widehat{\Phi}_{n,T} \leq x) < \mathbb{P}(\Phi_{n,T} \leq x)$ , it holds that

$$\begin{aligned} \left| \mathbb{P}(\widehat{\Phi}_{n,T} \leq x) - \mathbb{P}(\Phi_{n,T} \leq x) \right| &\leq \mathbb{P}(\Phi_{n,T} \leq x - \delta_T) - \mathbb{P}(\widehat{\Phi}_{n,T} \leq x - \delta_T) \\ &\quad + \mathbb{P}(\Phi_{n,T} \leq x) - \mathbb{P}(\Phi_{n,T} \leq x - \delta_T) \\ &\quad + \mathbb{P}(|\widehat{\Phi}_{n,T} - \Phi_{n,T}| > \delta_T). \end{aligned}$$

Consequently,

$$\begin{aligned} \sup_{x \in \mathbb{R}} \left| \mathbb{P}(\widehat{\Phi}_{n,T} \leq x) - \mathbb{P}(\Phi_{n,T} \leq x) \right| &\leq \sup_{x \in \mathbb{R}} \left| \mathbb{P}(\widehat{\Phi}_{n,T} \leq x) - \mathbb{P}(\Phi_{n,T} \leq x) \right| \\ &\quad + \sup_{x \in \mathbb{R}} \mathbb{P}(|\Phi_{n,T} - x| \leq \delta_T) \\ &\quad + \mathbb{P}(|\widehat{\Phi}_{n,T} - \Phi_{n,T}| > \delta_T). \end{aligned}$$

Since the terms on the right-hand side are all  $o(1)$  by Steps 2–4, we arrive at (C.6).  $\square$

## Details on Steps 1–5

**Proof of Proposition C.7.** Consider the stationary process  $\mathcal{E}_i = \{\varepsilon_{it} : 1 \leq t \leq T\}$  for some fixed  $i \in \{1, \dots, n\}$ . By Theorem 2.1 and Corollary 2.1 in Berkes et al. (2014), the following strong approximation result holds true: On a richer probability space, there exist a standard Brownian motion  $\mathbb{B}_i$  and a sequence  $\{\tilde{\varepsilon}_{it} : t \in \mathbb{N}\}$  such that  $[\tilde{\varepsilon}_{i1}, \dots, \tilde{\varepsilon}_{iT}] \stackrel{\mathcal{D}}{=} [\varepsilon_{i1}, \dots, \varepsilon_{iT}]$  for each  $T$  and

$$\max_{1 \leq t \leq T} \left| \sum_{s=1}^t \tilde{\varepsilon}_{is} - \sigma_i \mathbb{B}_i(t) \right| = o(T^{1/q}) \quad \text{a.s.}, \quad (\text{C.8})$$

where  $\sigma_i^2 = \sum_{k \in \mathbb{Z}} \text{Cov}(\varepsilon_{i0}, \varepsilon_{ik})$  denotes the long-run error variance. We apply this result separately for each  $i \in \{1, \dots, n\}$ . Since the error processes  $\mathcal{E}_i = \{\varepsilon_{it} : 1 \leq t \leq T\}$  are independent across  $i$ , we can construct the processes  $\tilde{\mathcal{E}}_i = \{\tilde{\varepsilon}_{it} : t \in \mathbb{N}\}$  in such a way that they are independent across  $i$  as well.

We now define the statistic  $\tilde{\Phi}_{n,T}$  in the same way as  $\hat{\Phi}_{n,T}$  except that the error processes  $\mathcal{E}_i$  are replaced by  $\tilde{\mathcal{E}}_i$ . Specifically, we set

$$\tilde{\Phi}_{n,T} = \max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} \left\{ \left| \frac{\tilde{\phi}_{ij,T}(u,h)}{(\tilde{\sigma}_i^2 + \tilde{\sigma}_j^2)^{1/2}} \right| - \lambda(h) \right\},$$

where

$$\tilde{\phi}_{ij,T}(u,h) = \sum_{t=1}^T w_{t,T}(u,h) \{(\tilde{\varepsilon}_{it} - \tilde{\varepsilon}_i) - (\tilde{\varepsilon}_{jt} - \tilde{\varepsilon}_j)\}$$

and the estimator  $\tilde{\sigma}_i^2$  is constructed from the sample  $\tilde{\mathcal{E}}_i$  in the same way as  $\hat{\sigma}_i^2$  is constructed from  $\mathcal{E}_i$ . Since  $[\tilde{\varepsilon}_{i1}, \dots, \tilde{\varepsilon}_{iT}] \stackrel{\mathcal{D}}{=} [\varepsilon_{i1}, \dots, \varepsilon_{iT}]$  and  $\hat{\sigma}_i^2 = \sigma_i^2 + o_p(\rho_T)$ , we have that  $\tilde{\sigma}_i^2 = \sigma_i^2 + o_p(\rho_T)$  as well. In addition to  $\tilde{\Phi}_{n,T}$ , we introduce the Gaussian statistic

$$\Phi_{n,T} = \max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} \left\{ \left| \frac{\phi_{ij,T}(u,h)}{(\sigma_i^2 + \sigma_j^2)^{1/2}} \right| - \lambda(h) \right\}$$

and the auxiliary statistic

$$\Phi_{n,T}^\diamond = \max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} \left\{ \left| \frac{\phi_{ij,T}(u,h)}{(\tilde{\sigma}_i^2 + \tilde{\sigma}_j^2)^{1/2}} \right| - \lambda(h) \right\},$$

where  $\phi_{ij,T}(u,h) = \sum_{t=1}^T w_{t,T}(u,h) \{\sigma_i(Z_{it} - \bar{Z}_i) - \sigma_j(Z_{jt} - \bar{Z}_j)\}$  and the Gaussian variables

$Z_{it}$  are chosen as  $Z_{it} = \mathbb{B}_i(t) - \mathbb{B}_i(t-1)$ . With this notation, we obtain the obvious bound

$$|\tilde{\Phi}_{n,T} - \Phi_{n,T}| \leq |\tilde{\Phi}_{n,T} - \Phi_{n,T}^\diamond| + |\Phi_{n,T}^\diamond - \Phi_{n,T}|.$$

In what follows, we prove that

$$|\tilde{\Phi}_{n,T} - \Phi_{n,T}^\diamond| = o_p\left(\frac{T^{1/q}}{\sqrt{Th_{\min}}}\right) \quad (\text{C.9})$$

$$|\Phi_{n,T}^\diamond - \Phi_{n,T}| = o_p(\rho_T \sqrt{\log T}), \quad (\text{C.10})$$

which completes the proof.

First consider  $|\tilde{\Phi}_{n,T} - \Phi_{n,T}^\diamond|$ . Straightforward calculations yield that

$$\begin{aligned} |\tilde{\Phi}_{n,T} - \Phi_{n,T}^\diamond| &\leq \max_{1 \leq i < j \leq n} (\tilde{\sigma}_i^2 + \tilde{\sigma}_j^2)^{-1/2} \max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} |\tilde{\phi}_{ij,T}(u,h) - \phi_{ij,T}(u,h)| \\ &= O_p(1) \cdot \max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} |\tilde{\phi}_{ij,T}(u,h) - \phi_{ij,T}(u,h)|, \end{aligned} \quad (\text{C.11})$$

where the last line follows from the fact that  $\tilde{\sigma}_i^2 = \sigma_i^2 + o_p(\rho_T)$ . Using summation by parts (that is,  $\sum_{t=1}^T a_t b_t = \sum_{t=1}^{T-1} A_t(b_t - b_{t+1}) + A_T b_T$  with  $A_t = \sum_{s=1}^t a_s$ ), we further obtain that

$$\begin{aligned} &|\tilde{\phi}_{ij,T}(u,h) - \phi_{ij,T}(u,h)| \\ &= \left| \sum_{t=1}^T w_{t,T}(u,h) \{(\tilde{\varepsilon}_{it} - \tilde{\varepsilon}_i) - (\tilde{\varepsilon}_{jt} - \tilde{\varepsilon}_j) - \sigma_i(Z_{it} - \bar{Z}_i) + \sigma_j(Z_{jt} - \bar{Z}_j)\} \right| \\ &= \left| \sum_{t=1}^{T-1} A_{ij,t}(w_{t,T}(u,h) - w_{t+1,T}(u,h)) + A_{ij,T} w_{T,T}(u,h) \right|, \end{aligned}$$

where

$$A_{ij,t} = \sum_{s=1}^t \{(\tilde{\varepsilon}_{is} - \tilde{\varepsilon}_i) - (\tilde{\varepsilon}_{js} - \tilde{\varepsilon}_j) - \sigma_i(Z_{is} - \bar{Z}_i) + \sigma_j(Z_{js} - \bar{Z}_j)\}$$

and  $A_{ij,T} = 0$  for all pairs  $(i,j)$  by construction. From this, it follows that

$$|\tilde{\phi}_{ij,T}(u,h) - \phi_{ij,T}(u,h)| \leq W_T(u,h) \max_{1 \leq t \leq T} |A_{ij,t}| \quad (\text{C.12})$$

with  $W_T(u,h) = \sum_{t=1}^{T-1} |w_{t+1,T}(u,h) - w_{t,T}(u,h)|$ . Straightforward calculations yield that

$$\begin{aligned} \max_{1 \leq t \leq T} |A_{ij,t}| &\leq \max_{1 \leq t \leq T} \left| \sum_{s=1}^t \tilde{\varepsilon}_{is} - \sigma_i \sum_{s=1}^t Z_{is} \right| + \max_{1 \leq t \leq T} |t(\tilde{\varepsilon}_i - \sigma_i \bar{Z}_i)| \\ &\quad + \max_{1 \leq t \leq T} \left| \sum_{s=1}^t \tilde{\varepsilon}_{js} - \sigma_j \sum_{s=1}^t Z_{js} \right| + \max_{1 \leq t \leq T} |t(\tilde{\varepsilon}_j - \sigma_j \bar{Z}_j)| \end{aligned}$$

$$\begin{aligned}
&\leq 2 \max_{1 \leq t \leq T} \left| \sum_{s=1}^t \tilde{\varepsilon}_{is} - \sigma_i \sum_{s=1}^t Z_{is} \right| + 2 \max_{1 \leq t \leq T} \left| \sum_{s=1}^t \tilde{\varepsilon}_{js} - \sigma_j \sum_{s=1}^t Z_{js} \right| \\
&= 2 \max_{1 \leq t \leq T} \left| \sum_{s=1}^t \tilde{\varepsilon}_{is} - \sigma_i \sum_{s=1}^t (\mathbb{B}_i(s) - \mathbb{B}_i(s-1)) \right| \\
&\quad + 2 \max_{1 \leq t \leq T} \left| \sum_{s=1}^t \tilde{\varepsilon}_{js} - \sigma_j \sum_{s=1}^t (\mathbb{B}_j(s) - \mathbb{B}_j(s-1)) \right| \\
&= 2 \max_{1 \leq t \leq T} \left| \sum_{s=1}^t \tilde{\varepsilon}_{is} - \sigma_i \mathbb{B}_i(t) \right| + 2 \max_{1 \leq t \leq T} \left| \sum_{s=1}^t \tilde{\varepsilon}_{js} - \sigma_j \mathbb{B}_j(t) \right|.
\end{aligned}$$

Applying the strong approximation result (C.8), we can infer that

$$\max_{1 \leq t \leq T} |A_{ij,t}| = o_p(T^{1/q}).$$

Moreover, standard arguments show that  $\max_{(u,h) \in \mathcal{G}_T} W_T(u,h) = O(1/\sqrt{Th_{\min}})$ . Plugging these two results into (C.12), we obtain that

$$\max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} |\tilde{\phi}_{ij,T}(u,h) - \phi_{ij,T}(u,h)| = o_p\left(\frac{T^{1/q}}{\sqrt{Th_{\min}}}\right),$$

which in view of (C.11) yields that  $|\tilde{\Phi}_{n,T} - \Phi_{n,T}^\diamond| = o_p(T^{1/q}/\sqrt{Th_{\min}})$ . This completes the proof of (C.9).

Next consider  $|\Phi_{n,T}^\diamond - \Phi_{n,T}|$ . It holds that

$$\begin{aligned}
|\Phi_{n,T}^\diamond - \Phi_{n,T}| &\leq \max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} \left| \frac{\phi_{ij,T}(u,h)}{\{\tilde{\sigma}_i^2 + \tilde{\sigma}_j^2\}^{1/2}} - \frac{\phi_{ij,T}(u,h)}{\{\sigma_i^2 + \sigma_j^2\}^{1/2}} \right| \\
&\leq \max_{1 \leq i < j \leq n} \left\{ \left| (\tilde{\sigma}_i^2 + \tilde{\sigma}_j^2)^{-1/2} - (\sigma_i^2 + \sigma_j^2)^{-1/2} \right| \right\} \max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} |\phi_{ij,T}(u,h)| \\
&= o_p(\rho_T) \max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} |\phi_{ij,T}(u,h)|, \tag{C.13}
\end{aligned}$$

where the last line is due to the fact that  $\tilde{\sigma}_i^2 = \sigma_i^2 + o_p(\rho_T)$ . We can write  $\phi_{ij,T}(u,h) = \phi_{ij,T}^{(I)}(u,h) - \phi_{ij,T}^{(II)}(u,h)$ , where

$$\begin{aligned}
\phi_{ij,T}^{(I)}(u,h) &= \sum_{t=1}^T w_{t,T}(u,h) (\sigma_i Z_{it} - \sigma_j Z_{jt}) \sim N(0, \sigma_i^2 + \sigma_j^2) \\
\phi_{ij,T}^{(II)}(u,h) &= \sum_{t=1}^T w_{t,T}(u,h) (\sigma_i \bar{Z}_i - \sigma_j \bar{Z}_j) \sim N(0, (\sigma_i^2 + \sigma_j^2) c_T(u,h))
\end{aligned}$$

with  $c_T(u,h) = \{\sum_{t=1}^T w_{t,T}(u,h)\}^2/T \leq C < \infty$  for all  $(u,h) \in \mathcal{G}_T$  and  $1 \leq i < j \leq n$ . This shows that  $\phi_{ij,T}(u,h)$  are centred Gaussian random variables with bounded variance for all

$(u, h) \in \mathcal{G}_T$  and  $1 \leq i < j \leq n$ . Hence, standard results on the maximum of Gaussian random variables yield that

$$\max_{1 \leq i < j \leq n} \max_{(u, h) \in \mathcal{G}_T} |\phi_{ij,T}(u, h)| = O_p(\sqrt{\log T}), \quad (\text{C.14})$$

where we have used that  $n$  is fixed and  $|\mathcal{G}_T| = O(T^\theta)$  for some large but fixed constant  $\theta$  by Assumption (C10). Plugging this into (C.13) yields  $|\Phi_{n,T}^\diamond - \Phi_{n,T}| = o_p(\rho_T \sqrt{\log T})$ , which completes the proof of (C.10).  $\square$

**Proof of Proposition C.8.** The proof is an application of anti-concentration bounds for Gaussian random vectors. We in particular make use of the following anti-concentration inequality from Nazarov (2003), which can also be found as Lemma A.1 in Chernozhukov et al. (2017).

**Lemma C.10.** *Let  $\mathbf{Z} = (Z_1, \dots, Z_p)^\top$  be a centred Gaussian random vector in  $\mathbb{R}^p$  such that  $\mathbb{E}[Z_j^2] \geq b$  for all  $1 \leq j \leq p$  and some constant  $b > 0$ . Then for every  $\mathbf{z} \in \mathbb{R}^p$  and  $a > 0$ ,*

$$\mathbb{P}(\mathbf{Z} \leq \mathbf{z} + a) - \mathbb{P}(\mathbf{Z} \leq \mathbf{z}) \leq Ca\sqrt{\log p},$$

where the constant  $C$  only depends on  $b$ .

To apply this result, we introduce the following notation: We write  $x = (u, h)$  and  $\mathcal{G}_T = \{x_1, \dots, x_p\}$ , where  $p := |\mathcal{G}_T| \leq O(T^\theta)$  for some large but fixed  $\theta > 0$  by our assumptions.

For  $k = 1, \dots, p$  and  $1 \leq i < j \leq n$ , we further let

$$Z_{ij,2k-1} = \frac{\phi_{ij,T}(x_{k1}, x_{k2})}{\{\sigma_i^2 + \sigma_j^2\}^{1/2}} \quad \text{and} \quad Z_{ij,2k} = -\frac{\phi_{ij,T}(x_{k1}, x_{k2})}{\{\sigma_i^2 + \sigma_j^2\}^{1/2}}$$

along with  $\lambda_{ij,2k-1} = \lambda(x_{k2})$  and  $\lambda_{ij,2k} = \lambda(x_{k2})$ , where  $x_k = (x_{k1}, x_{k2})$ . Under our assumptions, it holds that  $\mathbb{E}[Z_{ij,l}] = 0$  and  $\mathbb{E}[Z_{ij,l}^2] \geq b > 0$  for all  $i, j$  and  $l$ . We next construct the random vector  $\mathbf{Z} = (Z_{ij,l} : 1 \leq i < j \leq n, 1 \leq l \leq 2p)$  by stacking the variables  $Z_{ij,l}$  in a certain order (which can be chosen freely) and construct the vector  $\boldsymbol{\lambda} = (\lambda_{ij,l} : 1 \leq i < j \leq n, 1 \leq l \leq 2p)$  in an analogous way. Since the variables  $Z_{ij,l}$  are

normally distributed,  $\mathbf{Z}$  is a Gaussian random vector of length  $(n-1)np$ .

With this notation at hand, we can express the probability  $\mathbb{P}(\Phi_{n,T} \leq q)$  as follows for each  $q \in \mathbb{R}$ :

$$\begin{aligned}\mathbb{P}(\Phi_{n,T} \leq q) &= \mathbb{P}\left(\max_{1 \leq i < j \leq n} \max_{1 \leq l \leq 2p} \{Z_{ij,l} - \lambda_{ij,l}\} \leq q\right) \\ &= \mathbb{P}(Z_{ij,l} \leq \lambda_{ij,l} + q \text{ for all } (i, j, l)) = \mathbb{P}(\mathbf{Z} \leq \boldsymbol{\lambda} + q).\end{aligned}$$

Consequently,

$$\begin{aligned}\mathbb{P}(|\Phi_{n,T} - x| \leq \delta_T) &= \mathbb{P}(x - \delta_T \leq \Phi_{n,T} \leq x + \delta_T) \\ &= \mathbb{P}(\Phi_{n,T} \leq x + \delta_T) - \mathbb{P}(\Phi_{n,T} \leq x) \\ &\quad + \mathbb{P}(\Phi_{n,T} \leq x) - \mathbb{P}(\Phi_{n,T} \leq x - \delta_T) \\ &= \mathbb{P}(\mathbf{Z} \leq \boldsymbol{\lambda} + x + \delta_T) - \mathbb{P}(\mathbf{Z} \leq \boldsymbol{\lambda} + x) \\ &\quad + \mathbb{P}(\mathbf{Z} \leq \boldsymbol{\lambda} + x) - \mathbb{P}(\mathbf{Z} \leq \boldsymbol{\lambda} + x - \delta_T) \\ &\leq 2C\delta_T \sqrt{\log((n-1)np)},\end{aligned}$$

where the last line is by Lemma C.10. This immediately implies Proposition C.8.  $\square$

**Proof of Proposition C.9.** Straightforward calculations yield that

$$\begin{aligned}|\widehat{\Phi}_{n,T} - \widehat{\Phi}_{n,T}| &\leq \max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} \left| \frac{\widehat{\phi}_{ij,T}(u,h)}{(\widehat{\sigma}_i^2 + \widehat{\sigma}_j^2)^{1/2}} - \frac{\widehat{\phi}_{ij,T}(u,h)}{(\widehat{\sigma}_i^2 + \widehat{\sigma}_j^2)^{1/2}} \right| \\ &\quad + \max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} \left| \frac{\widehat{\phi}_{ij,T}(u,h)}{(\widehat{\sigma}_i^2 + \widehat{\sigma}_j^2)^{1/2}} - \frac{\widehat{\phi}_{ij,T}(u,h)}{(\widehat{\sigma}_i^2 + \widehat{\sigma}_j^2)^{1/2}} \right|.\end{aligned}$$

Since  $\widehat{\sigma}_i^2 = \sigma_i^2 + o_p(\rho_T)$  and  $\widehat{\sigma}_j^2 = \sigma_j^2 + o_p(\rho_T)$ , we further get that

$$\begin{aligned}&\max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} \left| \frac{\widehat{\phi}_{ij,T}(u,h)}{(\widehat{\sigma}_i^2 + \widehat{\sigma}_j^2)^{1/2}} - \frac{\widehat{\phi}_{ij,T}(u,h)}{(\widehat{\sigma}_i^2 + \widehat{\sigma}_j^2)^{1/2}} \right| \\ &\leq \max_{1 \leq i < j \leq n} \left\{ \left| (\widehat{\sigma}_i^2 + \widehat{\sigma}_j^2)^{-1/2} - (\sigma_i^2 + \sigma_j^2)^{-1/2} \right| \right\} \max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} |\widehat{\phi}_{ij,T}(u,h)| \\ &= o_p(\rho_T) \max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} |\widehat{\phi}_{ij,T}(u,h)|\end{aligned}$$

and

$$\begin{aligned}
& \max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} \left| \frac{\widehat{\phi}_{ij,T}(u,h)}{(\widehat{\sigma}_i^2 + \widehat{\sigma}_j^2)^{1/2}} - \frac{\widehat{\phi}_{ij,T}(u,h)}{(\widehat{\sigma}_i^2 + \widehat{\sigma}_j^2)^{1/2}} \right| \\
& \leq \max_{1 \leq i < j \leq n} \left\{ (\widehat{\sigma}_i^2 + \widehat{\sigma}_j^2)^{-1/2} \right\} \max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} \left| \widehat{\phi}_{ij,T}(u,h) - \widehat{\phi}_{ij,T}(u,h) \right| \\
& = O_p(1) \max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} \left| \widehat{\phi}_{ij,T}(u,h) - \widehat{\phi}_{ij,T}(u,h) \right|,
\end{aligned}$$

where the difference of the kernel averages  $\widehat{\phi}_{ij,T}(u,h) - \widehat{\phi}_{ij,T}(u,h)$  does not include the error terms (they cancel out) and can be written as

$$\begin{aligned}
& \left| \widehat{\phi}_{ij,T}(u,h) - \widehat{\phi}_{ij,T}(u,h) \right| \\
& = \left| \sum_{t=1}^T w_{t,T}(u,h) \{ (\beta_i - \widehat{\beta}_i)^\top (\mathbf{X}_{it} - \bar{\mathbf{X}}_i) - (\beta_j - \widehat{\beta}_j)^\top (\mathbf{X}_{jt} - \bar{\mathbf{X}}_j) \} \right| \\
& \leq \left| (\beta_i - \widehat{\beta}_i)^\top \sum_{t=1}^T w_{t,T}(u,h) \mathbf{X}_{it} \right| + \left| (\beta_i - \widehat{\beta}_i)^\top \bar{\mathbf{X}}_i \right| \left| \sum_{t=1}^T w_{t,T}(u,h) \right| \\
& \quad + \left| (\beta_j - \widehat{\beta}_j)^\top \sum_{t=1}^T w_{t,T}(u,h) \mathbf{X}_{jt} \right| + \left| (\beta_j - \widehat{\beta}_j)^\top \bar{\mathbf{X}}_j \right| \left| \sum_{t=1}^T w_{t,T}(u,h) \right|.
\end{aligned}$$

Hence,

$$\left| \widehat{\Phi}_{n,T} - \widehat{\Phi}_{n,T} \right| \leq o_p(\rho_T) A_{n,T} + O_p(1) \{ 2B_{n,T} + 2C_{n,T} \}, \quad (\text{C.15})$$

where

$$\begin{aligned}
A_{n,T} &= \max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} \left| \widehat{\phi}_{ij,T}(u,h) \right| \\
B_{n,T} &= \max_{1 \leq i \leq n} \max_{(u,h) \in \mathcal{G}_T} \left| (\beta_i - \widehat{\beta}_i)^\top \sum_{t=1}^T w_{t,T}(u,h) \mathbf{X}_{it} \right| \\
C_{n,T} &= \max_{1 \leq i \leq n} \left| (\beta_i - \widehat{\beta}_i)^\top \bar{\mathbf{X}}_i \right| \max_{(u,h) \in \mathcal{G}_T} \left| \sum_{t=1}^T w_{t,T}(u,h) \right|.
\end{aligned}$$

We examine these three terms separately.

We first prove that

$$A_{n,T} = \max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} \left| \widehat{\phi}_{ij,T}(u,h) \right| = O_p(\sqrt{\log T}). \quad (\text{C.16})$$

From the proof of Proposition C.7, we know that there exist identically distributed versions

$\tilde{\phi}_{ij,T}(u, h)$  of the statistics  $\widehat{\phi}_{ij,T}(u, h)$  with the property that

$$\max_{1 \leq i < j \leq n} \max_{(u, h) \in \mathcal{G}_T} |\tilde{\phi}_{ij,T}(u, h) - \phi_{ij,T}(u, h)| = o_p\left(\frac{T^{1/q}}{\sqrt{Th_{\min}}}\right). \quad (\text{C.17})$$

Instead of (C.16), it thus suffices to show that

$$\max_{1 \leq i < j \leq n} \max_{(u, h) \in \mathcal{G}_T} |\tilde{\phi}_{ij,T}(u, h)| = O_p(\sqrt{\log T}). \quad (\text{C.18})$$

Since for any constant  $c > 0$ ,

$$\begin{aligned} & \mathbb{P}\left(\max_{i,j,(u,h)} |\phi_{ij,T}(u, h)| \leq \frac{c\sqrt{\log T}}{2}\right) \\ & \leq \mathbb{P}\left(\max_{i,j,(u,h)} |\tilde{\phi}_{ij,T}(u, h)| \leq c\sqrt{\log T}\right) \\ & \quad + \mathbb{P}\left(\left|\max_{i,j,(u,h)} |\tilde{\phi}_{ij,T}(u, h)| - \max_{i,j,(u,h)} |\phi_{ij,T}(u, h)|\right| > \frac{c\sqrt{\log T}}{2}\right) \\ & \leq \mathbb{P}\left(\max_{i,j,(u,h)} |\tilde{\phi}_{ij,T}(u, h)| \leq c\sqrt{\log T}\right) \\ & \quad + \mathbb{P}\left(\max_{i,j,(u,h)} |\tilde{\phi}_{ij,T}(u, h) - \phi_{ij,T}(u, h)| > \frac{c\sqrt{\log T}}{2}\right) \end{aligned}$$

and  $\mathbb{P}(\max_{i,j,(u,h)} |\tilde{\phi}_{ij,T}(u, h) - \phi_{ij,T}(u, h)| > c\sqrt{\log T}/2) = o(1)$  by (C.17), we get that

$$\begin{aligned} & \mathbb{P}\left(\max_{i,j,(u,h)} |\tilde{\phi}_{ij,T}(u, h)| \leq c\sqrt{\log T}\right) \\ & \geq \mathbb{P}\left(\max_{i,j,(u,h)} |\phi_{ij,T}(u, h)| \leq \frac{c\sqrt{\log T}}{2}\right) - o(1). \end{aligned} \quad (\text{C.19})$$

Moreover, since  $\max_{i,j,(u,h)} |\phi_{ij,T}(u, h)| = O_p(\sqrt{\log T})$  as already proven in (C.14), we can make the probability  $\mathbb{P}(\max_{i,j,(u,h)} |\phi_{ij,T}(u, h)| \leq c\sqrt{\log T}/2)$  on the right-hand side of (C.19) arbitrarily close to 1 by choosing the constant  $c$  sufficiently large. Hence, for any  $\delta > 0$ , we can find a constant  $c > 0$  such that  $\mathbb{P}(\max_{i,j,(u,h)} |\tilde{\phi}_{ij,T}(u, h)| \leq c\sqrt{\log T}) \geq 1 - \delta$  for sufficiently large  $T$ . This proves (C.18), which in turn yields (C.16).

We next turn to  $B_{n,T}$ . Without loss of generality, we assume that  $\mathbf{X}_{it}$  is real-valued. The vector-valued case can be handled analogously. To start with, we have a closer look at the term  $\sum_{t=1}^T w_{t,T}(u, h) \mathbf{X}_{it}$ . By construction, the kernel weights  $w_{t,T}(u, h)$  are unequal to 0 if



and only if  $T(u - h) \leq t \leq T(u + h)$ . We can use this fact to write

$$\left| \sum_{t=1}^T w_{t,T}(u, h) \mathbf{X}_{it} \right| = \left| \sum_{t=\lfloor T(u-h) \rfloor}^{\lceil T(u+h) \rceil} w_{t,T}(u, h) \mathbf{X}_{it} \right|.$$

Note that

$$\sum_{t=\lfloor T(u-h) \rfloor}^{\lceil T(u+h) \rceil} w_{t,T}^2(u, h) = \sum_{t=1}^T w_{t,T}^2(u, h) = \sum_{t=1}^T \frac{\Lambda_{t,T}^2(u, h)}{\sum_{s=1}^T \Lambda_{s,T}^2(u, h)} = 1. \quad (\text{C.20})$$

Denoting by  $D_{T,u,h}$  the number of integers between  $\lfloor T(u - h) \rfloor$  and  $\lceil T(u + h) \rceil$  (with the obvious bounds  $2Th \leq D_{T,u,h} \leq 2Th + 2$ ) and using (C.20), we can normalize the kernel weights as follows:

$$\sum_{t=\lfloor T(u-h) \rfloor}^{\lceil T(u+h) \rceil} (\sqrt{D_{T,u,h}} \cdot w_{t,T}(u, h))^2 = D_{T,u,h}.$$

Next, we apply Proposition C.2 with the weights  $a_t = \sqrt{D_{T,u,h}} \cdot w_{t,T}(u, h)$  to obtain that

$$\begin{aligned} \mathbb{P} \left( \left| \sum_{t=\lfloor T(u-h) \rfloor}^{\lceil T(u+h) \rceil} \sqrt{D_{T,u,h}} \cdot w_{t,T}(u, h) \mathbf{X}_{it} \right| \geq x \right) \\ \leq C_1 \frac{(\sum_{t=\lfloor T(u-h) \rfloor}^{\lceil T(u+h) \rceil} |\sqrt{D_{T,u,h}} \cdot w_{t,T}(u, h)|^{q'}) \|\mathbf{X}_i\|_{q',\alpha}^{q'}}{x^{q'}} \\ + C_2 \exp \left( -\frac{C_3 x^2}{D_{T,u,h} \|\mathbf{X}_i\|_{2,\alpha}^2} \right) \end{aligned} \quad (\text{C.21})$$

for any  $x > 0$ , where  $\|\mathbf{X}_i\|_{q',\alpha}^{q'} = \sup_{t \geq 0} (t+1)^\alpha \sum_{s=t}^\infty \delta_{q'}(\mathbf{h}_i, s)$  is the dependence adjusted norm introduced in Definition C.1 and  $\|\mathbf{X}_i\|_{q',\alpha}^{q'} < \infty$  by Assumption (C6). From (C.21), it follows that for any  $\delta > 0$ ,

$$\begin{aligned} \mathbb{P} \left( \max_{(u,h) \in \mathcal{G}_T} \left| \sum_{t=\lfloor T(u-h) \rfloor}^{\lceil T(u+h) \rceil} w_{t,T}(u, h) \mathbf{X}_{it} \right| \geq \delta T^{1/q} \right) \\ \leq \sum_{(u,h) \in \mathcal{G}_T} \mathbb{P} \left( \left| \sum_{t=\lfloor T(u-h) \rfloor}^{\lceil T(u+h) \rceil} w_{t,T}(u, h) \mathbf{X}_{it} \right| \geq \delta T^{1/q} \right) \\ = \sum_{(u,h) \in \mathcal{G}_T} \mathbb{P} \left( \left| \sum_{t=\lfloor T(u-h) \rfloor}^{\lceil T(u+h) \rceil} \sqrt{D_{T,u,h}} \cdot w_{t,T}(u, h) \mathbf{X}_{it} \right| \geq \delta \sqrt{D_{T,u,h}} T^{1/q} \right) \\ \leq \sum_{(u,h) \in \mathcal{G}_T} \left[ C_1 \frac{(\sqrt{D_{T,u,h}})^{q'} (\sum |w_{t,T}(u, h)|^{q'}) \|\mathbf{X}_i\|_{q',\alpha}^{q'}}{(\delta \sqrt{D_{T,u,h}} T^{1/q})^{q'}} + C_2 \exp \left( -\frac{C_3 (\delta \sqrt{D_{T,u,h}} T^{1/q})^2}{D_{T,u,h} \|\mathbf{X}_i\|_{2,\alpha}^2} \right) \right] \end{aligned}$$

$$\begin{aligned}
&= \sum_{(u,h) \in \mathcal{G}_T} \left[ C_1 \frac{(\sum |w_{t,T}(u,h)|^{q'}) \|\mathbf{X}_{i\cdot}\|_{q',\alpha}^{q'}}{\delta^{q'} T^{q'/q}} + C_2 \exp\left(-\frac{C_3 \delta^2 T^{2/q}}{\|\mathbf{X}_{i\cdot}\|_{2,\alpha}^2}\right) \right] \\
&\leq C_1 \frac{T^\theta \|\mathbf{X}_{i\cdot}\|_{q',\alpha}^{q'}}{\delta^{q'} T^{q'/q}} \max_{(u,h) \in \mathcal{G}_T} \left( \sum_{t=\lfloor T(u-h) \rfloor}^{\lceil T(u+h) \rceil} |w_{t,T}(u,h)|^{q'} \right) + C_2 T^\theta \exp\left(-\frac{C_3 \delta^2 T^{2/q}}{\|\mathbf{X}_{i\cdot}\|_{2,\alpha}^2}\right) \\
&= C \frac{T^{\theta-q'/q}}{\delta^{q'}} + C T^\theta \exp(-C T^{2/q} \delta^2), \tag{C.22}
\end{aligned}$$

where the constant  $C$  depends neither on  $T$  nor on  $\delta$ . In the last equality of the above display, we have used the following facts:

- (i)  $\|\mathbf{X}_{i\cdot}\|_{q',\alpha}^{q'} < \infty$  by Assumption (C6).
- (ii)  $\|\mathbf{X}_{i\cdot}\|_{2,\alpha}^2 < \infty$  (which follows from (i)).
- (iii)  $\max_{(u,h) \in \mathcal{G}_T} (\sum_{t=\lfloor T(u-h) \rfloor}^{\lceil T(u+h) \rceil} |w_{t,T}(u,h)|^{q'}) \leq 1$  for the following reason: By (C.20), it holds that  $\sum_{t=1}^T w_{t,T}^2(u,h) = 1$  and thus  $0 \leq w_{t,T}^2(u,h) \leq 1$  for all  $t, T$  and  $(u,h)$ . This implies that  $0 \leq |w_{t,T}(u,h)|^{q'} = (w_{t,T}^2(u,h))^{q'/2} \leq w_{t,T}^2(u,h) \leq 1$  for all  $t, T$  and  $(u,h)$ .

As a result,

$$\max_{(u,h) \in \mathcal{G}_T} \left( \sum_{t=\lfloor T(u-h) \rfloor}^{\lceil T(u+h) \rceil} |w_{t,T}(u,h)|^{q'} \right) \leq \max_{(u,h) \in \mathcal{G}_T} \left( \sum_{t=\lfloor T(u-h) \rfloor}^{\lceil T(u+h) \rceil} w_{t,T}^2(u,h) \right) = 1.$$

Since  $\theta - q'/q < 0$  by Assumption (C4), the bound in (C.22) converges to 0 as  $T \rightarrow \infty$  for any fixed  $\delta > 0$ . Consequently, we obtain that

$$\max_{(u,h) \in \mathcal{G}_T} \left| \sum_{t=\lfloor T(u-h) \rfloor}^{\lceil T(u+h) \rceil} w_{t,T}(u,h) \mathbf{X}_{it} \right| = o_p(T^{1/q}). \tag{C.23}$$

Using this together with the fact that  $\beta_i - \hat{\beta}_i = O_p(1/\sqrt{T})$  (which is the statement of Lemma C.5), we arrive at the bound

$$B_{n,T} = \max_{1 \leq i \leq n} \max_{(u,h) \in \mathcal{G}_T} \left| (\beta_i - \hat{\beta}_i)^\top \sum_{t=1}^T w_{t,T}(u,h) \mathbf{X}_{it} \right| = o_p\left(\frac{T^{1/q}}{\sqrt{T}}\right).$$

We finally turn to  $C_{n,T}$ . By straightforward calculations,  $|\sum_{t=1}^T w_{t,T}(u,h)| \leq C\sqrt{Th_{\max}} = o(\sqrt{T})$ . Moreover,  $\bar{\mathbf{X}}_i = O_p(1/\sqrt{T})$  by Lemma C.3 and  $\beta_i - \hat{\beta}_i = O_p(1/\sqrt{T})$  by Lemma

C.5. This immediately yields that

$$C_{n,T} = \max_{1 \leq i \leq n} |(\beta_i - \hat{\beta}_i)^\top \bar{\mathbf{X}}_i| \max_{(u,h) \in \mathcal{G}_T} \left| \sum_{t=1}^T w_{t,T}(u, h) \right| = o_p\left(\frac{1}{\sqrt{T}}\right).$$

To summarize, we have shown that

$$\begin{aligned} |\hat{\Phi}_{n,T} - \hat{\Phi}_{n,T}| &\leq o_p(\rho_T) A_{n,T} + O_p(1) \{2B_{n,T} + 2C_{n,T}\} \\ &= o_p(\rho_T) O_p(\sqrt{\log T}) + o_p\left(\frac{T^{1/q}}{\sqrt{T}}\right) + o_p\left(\frac{1}{\sqrt{T}}\right). \end{aligned}$$

This immediately implies the desired result.  $\square$

## Proof of Proposition 4.4

We first show that

$$\mathbb{P}(\Phi_{n,T} \leq q_{n,T}(\alpha)) = 1 - \alpha. \quad (\text{C.24})$$

We proceed by contradiction. Suppose that (C.24) does not hold true. Since  $\mathbb{P}(\Phi_{n,T} \leq q_{n,T}(\alpha)) \geq 1 - \alpha$  by definition of the quantile  $q_{n,T}(\alpha)$ , there exists  $\xi > 0$  such that  $\mathbb{P}(\Phi_{n,T} \leq q_{n,T}(\alpha)) = 1 - \alpha + \xi$ . From the proof of Proposition C.8, we know that for any  $\delta > 0$ ,

$$\begin{aligned} &\mathbb{P}(\Phi_{n,T} \leq q_{n,T}(\alpha)) - \mathbb{P}(\Phi_{n,T} \leq q_{n,T}(\alpha) - \delta) \\ &\leq \sup_{x \in \mathbb{R}} \mathbb{P}(|\Phi_{n,T} - x| \leq \delta) \leq 2C\delta\sqrt{\log((n-1)np)}. \end{aligned}$$

Hence,

$$\begin{aligned} \mathbb{P}(\Phi_{n,T} \leq q_{n,T}(\alpha) - \delta) &\geq \mathbb{P}(\Phi_{n,T} \leq q_{n,T}(\alpha)) - 2C\delta\sqrt{\log((n-1)np)} \\ &= 1 - \alpha + \xi - 2C\delta\sqrt{\log((n-1)np)} > 1 - \alpha \end{aligned}$$

for  $\delta > 0$  small enough. This contradicts the definition of the quantile  $q_{n,T}(\alpha)$  according to which  $q_{n,T}(\alpha) = \inf_{q \in \mathbb{R}} \{\mathbb{P}(\Phi_{n,T} \leq q) \geq 1 - \alpha\}$ . We thus arrive at (C.24).

Proposition 4.4 is a simple consequence of Theorem 4.1 and equation (C.24). Specifically, we obtain that under  $H_0$ ,

$$|\mathbb{P}(\hat{\Psi}_{n,T} \leq q_{n,T}(\alpha)) - (1 - \alpha)| = |\mathbb{P}(\hat{\Phi}_{n,T} \leq q_{n,T}(\alpha)) - (1 - \alpha)|$$

$$\begin{aligned}
&= \left| \mathbb{P}(\widehat{\Phi}_{n,T} \leq q_{n,T}(\alpha)) - \mathbb{P}(\Phi_{n,T} \leq q_{n,T}(\alpha)) \right| \\
&\leq \sup_{x \in \mathbb{R}} \left| \mathbb{P}(\widehat{\Phi}_{n,T} \leq x) - \mathbb{P}(\Phi_{n,T} \leq x) \right| = o(1).
\end{aligned}$$

## Proof of Proposition 4.5

To start with, note that for some sufficiently large constant  $C$  we have

$$\lambda(h) = \sqrt{2 \log\{1/(2h)\}} \leq \sqrt{2 \log\{1/(2h_{\min})\}} \leq C \sqrt{\log T}. \quad (\text{C.25})$$

Write  $\widehat{\psi}_{ij,T}(u, h) = \widehat{\psi}_{ij,T}^A(u, h) + \widehat{\psi}_{ij,T}^B(u, h)$  with

$$\begin{aligned}
\widehat{\psi}_{ij,T}^A(u, h) &= \sum_{t=1}^T w_{t,T}(u, h) \left\{ (\varepsilon_{it} - \bar{\varepsilon}_i) + (\beta_i - \widehat{\beta}_i)^\top (\mathbf{X}_{it} - \bar{\mathbf{X}}_i) - \bar{m}_{i,T} \right. \\
&\quad \left. - (\varepsilon_{jt} - \bar{\varepsilon}_j) - (\beta_j - \widehat{\beta}_j)^\top (\mathbf{X}_{jt} - \bar{\mathbf{X}}_j) + \bar{m}_{j,T} \right\} \\
\widehat{\psi}_{ij,T}^B(u, h) &= \sum_{t=1}^T w_{t,T}(u, h) \left( m_{i,T}\left(\frac{t}{T}\right) - m_{j,T}\left(\frac{t}{T}\right) \right),
\end{aligned}$$

where  $\bar{m}_{i,T} = T^{-1} \sum_{t=1}^T m_{i,T}(t/T)$ . Without loss of generality, consider the following scenario: there exists  $(u_0, h_0) \in \mathcal{G}_T$  with  $[u_0 - h_0, u_0 + h_0] \subseteq [0, 1]$  such that

$$m_{i,T}(w) - m_{j,T}(w) \geq c_T \sqrt{\log T / (Th_0)} \quad (\text{C.26})$$

for all  $w \in [u_0 - h_0, u_0 + h_0]$ .

We first derive a lower bound on the term  $\widehat{\psi}_{ij,T}^B(u_0, h_0)$ . Since the kernel  $K$  is symmetric and  $u_0 = t/T$  for some  $t$ , it holds that  $S_{T,1}(u_0, h_0) = 0$  and thus,

$$\begin{aligned}
w_{t,T}(u_0, h_0) &= \frac{K\left(\frac{\frac{t}{T} - u_0}{h_0}\right) S_{T,2}(u_0, h_0)}{\left\{ \sum_{t=1}^T K^2\left(\frac{\frac{t}{T} - u_0}{h_0}\right) S_{T,2}^2(u_0, h_0) \right\}^{1/2}} \\
&= \frac{K\left(\frac{\frac{t}{T} - u_0}{h_0}\right)}{\left\{ \sum_{t=1}^T K^2\left(\frac{\frac{t}{T} - u_0}{h_0}\right) \right\}^{1/2}} \geq 0.
\end{aligned}$$

Together with (C.26), this implies that

$$\widehat{\psi}_{ij,T}^B(u_0, h_0) \geq c_T \sqrt{\frac{\log T}{Th_0}} \sum_{t=1}^T w_{t,T}(u_0, h_0). \quad (\text{C.27})$$

Using the Lipschitz continuity of the kernel  $K$ , we can show by straightforward calculations

that for any  $(u, h) \in \mathcal{G}_T$  and any natural number  $\ell$ ,

$$\left| \frac{1}{Th} \sum_{t=1}^T K\left(\frac{\frac{t}{T} - u}{h}\right) \left(\frac{\frac{t}{T} - u}{h}\right)^\ell - \int_0^1 \frac{1}{h} K\left(\frac{w - u}{h}\right) \left(\frac{w - u}{h}\right)^\ell dw \right| \leq \frac{C}{Th}, \quad (\text{C.28})$$

where the constant  $C$  does not depend on  $u$ ,  $h$  and  $T$ . With the help of (C.28), we obtain that for any  $(u, h) \in \mathcal{G}_T$  with  $[u - h, u + h] \subseteq [0, 1]$ ,

$$\left| \sum_{t=1}^T w_{t,T}(u, h) - \frac{\sqrt{Th}}{\kappa} \right| \leq \frac{C}{\sqrt{Th}}, \quad (\text{C.29})$$

where  $\kappa = (\int K^2(\varphi) d\varphi)^{1/2}$  and the constant  $C$  does once again not depend on  $u$ ,  $h$  and  $T$ . From (C.29), it follows that  $\sum_{t=1}^T w_{t,T}(u, h) \geq \sqrt{Th}/(2\kappa)$  for sufficiently large  $T$  and any  $(u, h) \in \mathcal{G}_T$  with  $[u - h, u + h] \subseteq [0, 1]$ . This together with (C.27) allows us to infer that

$$\hat{\psi}_{ij,T}^B(u_0, h_0) \geq \frac{c_T \sqrt{\log T}}{2\kappa} \quad (\text{C.30})$$

for sufficiently large  $T$ .

We next analyze  $\hat{\psi}_{ij,T}^A(u_0, h_0)$ , which can be expressed as  $\hat{\psi}_{ij,T}^A(u_0, h_0) = \hat{\phi}_{ij,T}(u, h) + (\bar{m}_{j,T} - \bar{m}_{i,T}) \sum_{t=1}^T w_{t,T}(u, h)$ . The proof of Proposition C.9 shows that

$$\max_{1 \leq i < j \leq n} \max_{(u, h) \in \mathcal{G}_T} \left| \hat{\phi}_{ij,T}(u, h) \right| = O_p(\sqrt{\log T}).$$

Using this together with the bounds  $\bar{m}_{i,T} \leq C/T$  and  $\sum_{t=1}^T w_{t,T}(u, h) \leq C\sqrt{T}$ , we can infer that

$$\begin{aligned} & \max_{1 \leq i < j \leq n} \max_{(u, h) \in \mathcal{G}_T} \left| \hat{\psi}_{ij,T}^A(u, h) \right| \\ &= \max_{1 \leq i < j \leq n} \max_{(u, h) \in \mathcal{G}_T} \left| \hat{\phi}_{ij,T}(u, h) + (\bar{m}_{j,T} - \bar{m}_{i,T}) \sum_{t=1}^T w_{t,T}(u, h) \right| = O_p(\sqrt{\log T}). \end{aligned} \quad (\text{C.31})$$

With the help of (C.30), (C.31), (C.25) and the assumption that  $\hat{\sigma}_i^2 = \sigma_i^2 + o_p(\rho_T)$ , we finally arrive at

$$\begin{aligned} \hat{\Psi}_{n,T} &\geq \max_{1 \leq i < j \leq n} \max_{(u, h) \in \mathcal{G}_T} \frac{|\hat{\psi}_{ij,T}^B(u, h)|}{\{\hat{\sigma}_i^2 + \hat{\sigma}_j^2\}^{1/2}} - \max_{1 \leq i < j \leq n} \max_{(u, h) \in \mathcal{G}_T} \left\{ \frac{|\hat{\psi}_{ij,T}^A(u, h)|}{\{\hat{\sigma}_i^2 + \hat{\sigma}_j^2\}^{1/2}} + \lambda(h) \right\} \\ &= \max_{1 \leq i < j \leq n} \max_{(u, h) \in \mathcal{G}_T} \frac{|\hat{\psi}_{ij,T}^B(u, h)|}{\{\hat{\sigma}_i^2 + \hat{\sigma}_j^2\}^{1/2}} + O_p(\sqrt{\log T}) \\ &\geq \frac{c_T \sqrt{\log T}}{2\kappa} \min_{1 \leq i < j \leq n} \{\hat{\sigma}_i^2 + \hat{\sigma}_j^2\}^{-1/2} + O_p(\sqrt{\log T}) \end{aligned}$$

$$= \frac{c_T \sqrt{\log T}}{2\kappa} O_p(1) + O_p(\sqrt{\log T}). \quad (\text{C.32})$$

Since  $q_{n,T}(\alpha) = O(\sqrt{\log T})$  for any fixed  $\alpha \in (0, 1)$  and  $c_T \rightarrow \infty$ , (C.32) immediately implies that  $\mathbb{P}(\widehat{\Psi}_{n,T} \leq q_{n,T}(\alpha)) = o(1)$ .

## Proof of Proposition 4.6

Denote by  $\mathcal{M}_0$  the set of quadruples  $(i, j, u, h) \in \{1 \dots, n\}^2 \times \mathcal{G}_T$  for which  $H_0^{[i,j]}(u, h)$  is true. Then we can write the FWER as

$$\begin{aligned} \text{FWER}(\alpha) &= \mathbb{P}\left(\exists (i, j, u, h) \in \mathcal{M}_0 : \widehat{\psi}_{ij,T}^0(u, h) > q_{n,T}(\alpha)\right) \\ &= \mathbb{P}\left(\max_{(i,j,u,h) \in \mathcal{M}_0} \widehat{\psi}_{ij,T}^0(u, h) > q_{n,T}(\alpha)\right) \\ &= \mathbb{P}\left(\max_{(i,j,u,h) \in \mathcal{M}_0} \widehat{\phi}_{ij,T}^0(u, h) > q_{n,T}(\alpha)\right) \\ &\leq \mathbb{P}\left(\max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} \widehat{\phi}_{ij,T}^0(u, h) > q_{n,T}(\alpha)\right) \\ &= \mathbb{P}(\widehat{\Phi}_{n,T} > q_{n,T}(\alpha)) = \alpha + o(1), \end{aligned}$$

where the third equality uses that  $\widehat{\psi}_{ijk,T}^0 = \widehat{\phi}_{ijk,T}^0$  under  $H_0^{[i,j]}(u, h)$ .

## Proof of Proposition 5.1

For the sake of brevity, we introduce the following notation. For each  $i$  and  $j$ , we define the statistic  $\widehat{\Psi}_{ij,T} := \max_{(u,h) \in \mathcal{G}_T} \widehat{\psi}_{ij,T}^0(u, h)$  which can be interpreted as a distance measure between the two curves  $m_i$  and  $m_j$  on the whole interval  $[0, 1]$ . Using this notation, we can rewrite the dissimilarity measure defined in (5.1) as

$$\widehat{\Delta}(S, S') = \max_{\substack{i \in S, \\ j \in S'}} \widehat{\Psi}_{ij,T}.$$

Now consider the event

$$B_{n,T} = \left\{ \max_{1 \leq \ell \leq N} \max_{i,j \in G_\ell} \widehat{\Psi}_{ij,T} \leq q_{n,T}(\alpha) \text{ and } \min_{1 \leq \ell < \ell' \leq N} \min_{\substack{i \in G_\ell, \\ j \in G_{\ell'}}} \widehat{\Psi}_{ij,T} > q_{n,T}(\alpha) \right\}.$$

The term  $\max_{1 \leq \ell \leq N} \max_{i,j \in G_\ell} \widehat{\Psi}_{ij,T}$  is the largest multiscale distance between two time series  $i$  and  $j$  from the same group, whereas  $\min_{1 \leq \ell < \ell' \leq N} \min_{i \in G_\ell, j \in G_{\ell'}} \widehat{\Psi}_{ij,T}$  is the smallest

multiscale distance between two time series from two different groups. On the event  $B_{n,T}$ , it obviously holds that

$$\max_{1 \leq \ell \leq N} \max_{i,j \in G_\ell} \widehat{\Psi}_{ij,T} < \min_{1 \leq \ell < \ell' \leq N} \min_{\substack{i \in G_\ell, \\ j \in G_{\ell'}}} \widehat{\Psi}_{ij,T}. \quad (\text{C.33})$$

Hence, any two time series from the same class have a smaller distance than any two time series from two different classes. With the help of Proposition 4.4, it is easy to see that

$$\mathbb{P}\left(\max_{1 \leq \ell \leq N} \max_{i,j \in G_\ell} \widehat{\Psi}_{ij,T} \leq q_{n,T}(\alpha)\right) \geq (1 - \alpha) + o(1).$$

Moreover, the same arguments as for Proposition 4.5 show that

$$\mathbb{P}\left(\min_{1 \leq \ell < \ell' \leq N} \min_{\substack{i \in G_\ell, \\ j \in G_{\ell'}}} \widehat{\Psi}_{ij,T} \leq q_{n,T}(\alpha)\right) = o(1).$$

Taken together, these two statements imply that

$$\mathbb{P}(B_{n,T}) \geq (1 - \alpha) + o(1). \quad (\text{C.34})$$

In what follows, we show that on the event  $B_{n,T}$ , (i)  $\{\widehat{G}_1^{[n-N]}, \dots, \widehat{G}_N^{[n-N]}\} = \{G_1, \dots, G_N\}$  and (ii)  $\widehat{N} = N$ . From (i), (ii) and (C.34), the statements of Proposition 5.1 follow immediately.

**Proof of (i).** Suppose we are on the event  $B_{n,T}$ . The proof proceeds by induction on the iteration steps  $r$  of the HAC algorithm.

*Base case* ( $r = 0$ ): In the first iteration step, the HAC algorithm merges two singleton clusters  $\widehat{G}_i^{[0]} = \{i\}$  and  $\widehat{G}_j^{[0]} = \{j\}$  with  $i$  and  $j$  belonging to the same group  $G_k$ . This is a direct consequence of (C.33). The algorithm thus produces a partition  $\{\widehat{G}_1^{[1]}, \dots, \widehat{G}_{n-1}^{[1]}\}$  whose elements  $\widehat{G}_\ell^{[1]}$  all have the following property:  $\widehat{G}_\ell^{[1]} \subseteq G_k$  for some  $k$ , that is, each cluster  $\widehat{G}_\ell^{[1]}$  contains elements from only one group.

*Induction step* ( $r \rightsquigarrow r+1$ ): Now suppose we are in the  $r$ -th iteration step for some  $r < n - N$ . Assume that the partition  $\{\widehat{G}_1^{[r]}, \dots, \widehat{G}_{n-r}^{[r]}\}$  is such that for any  $\ell$ ,  $\widehat{G}_\ell^{[r]} \subseteq G_k$  for some  $k$ . Because of (C.33), the dissimilarity  $\widehat{\Delta}(\widehat{G}_\ell^{[r]}, \widehat{G}_{\ell'}^{[r]})$  gets minimal for two clusters  $\widehat{G}_\ell^{[r]}$  and  $\widehat{G}_{\ell'}^{[r]}$

with the property that  $\widehat{G}_\ell^{[r]} \cup \widehat{G}_{\ell'}^{[r]} \subseteq G_k$  for some  $k$ . Hence, the HAC algorithm produces a partition  $\{\widehat{G}_1^{[r+1]}, \dots, \widehat{G}_{n-(r+1)}^{[r+1]}\}$  whose elements  $\widehat{G}_\ell^{[r+1]}$  are all such that  $\widehat{G}_\ell^{[r+1]} \subseteq G_k$  for some  $k$ .

The above induction argument shows the following: For any  $r \leq n - N$ , the partition  $\{\widehat{G}_1^{[r]}, \dots, \widehat{G}_{n-r}^{[r]}\}$  consists of clusters  $\widehat{G}_\ell^{[r]}$  which all have the property that  $\widehat{G}_\ell^{[r]} \subseteq G_k$  for some  $k$ . This in particular holds for the partition  $\{\widehat{G}_1^{[n-N]}, \dots, \widehat{G}_N^{[n-N]}\}$ , which implies that  $\{\widehat{G}_1^{[n-N]}, \dots, \widehat{G}_N^{[n-N]}\} = \{G_1, \dots, G_N\}$ .  $\square$

**Proof of (ii).** To start with, consider any partition  $\{\widehat{G}_1^{[n-r]}, \dots, \widehat{G}_r^{[n-r]}\}$  with  $r < N$  elements. Such a partition must contain at least one element  $\widehat{G}_\ell^{[n-r]}$  with the following property:  $\widehat{G}_\ell^{[n-r]} \cap G_k \neq \emptyset$  and  $\widehat{G}_\ell^{[n-r]} \cap G_{k'} \neq \emptyset$  for some  $k \neq k'$ . On the event  $B_{n,T}$ , it obviously holds that  $\widehat{\Delta}(S) > q_{n,T}(\alpha)$  for any  $S$  with the property that  $S \cap G_k \neq \emptyset$  and  $S \cap G_{k'} \neq \emptyset$  for some  $k \neq k'$ . Hence, we can infer that on the event  $B_{n,T}$ ,  $\max_{1 \leq \ell \leq r} \widehat{\Delta}(\widehat{G}_\ell^{[n-r]}) > q_{n,T}(\alpha)$  for any  $r < N$ .

Next consider the partition  $\{\widehat{G}_1^{[n-r]}, \dots, \widehat{G}_r^{[n-r]}\}$  with  $r = N$  and suppose we are on the event  $B_{n,T}$ . From (i), we already know that  $\{\widehat{G}_1^{[n-N]}, \dots, \widehat{G}_N^{[n-N]}\} = \{G_1, \dots, G_N\}$ . Moreover,  $\widehat{\Delta}(G_\ell) \leq q_{n,T}(\alpha)$  for any  $\ell$ . Hence, we obtain that  $\max_{1 \leq \ell \leq N} \widehat{\Delta}(\widehat{G}_\ell^{[n-N]}) = \max_{1 \leq \ell \leq N} \widehat{\Delta}(G_\ell) \leq q_{n,T}(\alpha)$ .

Putting everything together, we can conclude that on the event  $B_{n,T}$ ,

$$\min \left\{ r = 1, 2, \dots \mid \max_{1 \leq \ell \leq r} \widehat{\Delta}(\widehat{G}_\ell^{[n-r]}) \leq q_{n,T}(\alpha) \right\} = N,$$

that is,  $\widehat{N} = N$ .  $\square$

## Proof of Proposition 5.2

We consider the event

$$D_{n,T} = \left\{ \widehat{\Phi}_{n,T} \leq q_{n,T}(\alpha) \text{ and } \min_{1 \leq \ell < \ell' \leq N} \min_{\substack{i \in G_\ell, \\ j \in G_{\ell'}}} \widehat{\Psi}_{ij,T} > q_{n,T}(\alpha) \right\},$$



where we write the statistic  $\widehat{\Phi}_{n,T}$  as

$$\widehat{\Phi}_{n,T} = \max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} \left\{ \left| \frac{\widehat{\psi}_{ij,T}(u,h) - \widehat{\psi}_{ij,T}^{\text{trend}}(u,h)}{(\widehat{\sigma}_i^2 + \widehat{\sigma}_j^2)^{1/2}} \right| - \lambda(h) \right\}$$

with  $\widehat{\psi}_{ij,T}^{\text{trend}}(u,h) = \sum_{t=1}^T w_{t,T}(u,h) \{(m_{i,T}(t/T) - \bar{m}_{i,T}) - (m_{j,T}(t/T) - \bar{m}_{j,T})\}$  and  $\bar{m}_{i,T} = T^{-1} \sum_{t=1}^T m_{i,T}(t/T)$ . The event  $D_{n,T}$  can be analysed by the same arguments as those applied to the event  $B_{n,T}$  in the proof of Proposition 5.1. In particular, analogous to (C.34) and statements (i) and (ii) in this proof, we can show that

$$\mathbb{P}(D_{n,T}) \geq (1 - \alpha) + o(1) \quad (\text{C.35})$$

and

$$D_{n,T} \subseteq \{\widehat{N} = N \text{ and } \widehat{G}_\ell = G_\ell \text{ for all } \ell\}. \quad (\text{C.36})$$

Moreover, we have that

$$D_{n,T} \subseteq \bigcap_{1 \leq \ell < \ell' \leq \widehat{N}} E_{n,T}^{[\ell, \ell']}(\alpha), \quad (\text{C.37})$$

which is a consequence of the following observation: For all  $i, j$  and  $(u, h) \in \mathcal{G}_T$  with

$$\left| \frac{\widehat{\psi}_{ij,T}(u,h) - \widehat{\psi}_{ij,T}^{\text{trend}}(u,h)}{(\widehat{\sigma}_i^2 + \widehat{\sigma}_j^2)^{1/2}} \right| - \lambda(h) \leq q_{n,T}(\alpha) \quad \text{and} \quad \left| \frac{\widehat{\psi}_{ij,T}(u,h)}{(\widehat{\sigma}_i^2 + \widehat{\sigma}_j^2)^{1/2}} \right| - \lambda(h) > q_{n,T}(\alpha),$$

it holds that  $\widehat{\psi}_{ij,T}^{\text{trend}}(u,h) \neq 0$ , which in turn implies that  $m_i(v) - m_j(v) \neq 0$  for some  $v \in I_{u,h}$ . From (C.36) and (C.37), we obtain that

$$D_{n,T} \subseteq \left\{ \bigcap_{1 \leq \ell < \ell' \leq \widehat{N}} E_{n,T}^{[\ell, \ell']}(\alpha) \right\} \cap \{\widehat{N} = N \text{ and } \widehat{G}_\ell = G_\ell \text{ for all } \ell\} = E_{n,T}(\alpha).$$

This together with (C.35) implies that  $\mathbb{P}(E_{n,T}(\alpha)) \geq (1 - \alpha) + o(1)$ .

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